Zurich University of the Arts

Game Design

Orientation C

Game Jam as Design Method

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Abstract

A game jam is a video-game development event, where game designers, programmers, artists and musicians come together to create a game in a limited period of time. It takes a process that usually requires years and compresses it into the duration of a single weekend.
As these events become more and more popular, it is now important, not only for the world of game design, but also for the world of design, to understand game jams and how they can be used in different contexts and in different shapes.
The goal of this master thesis is to define and to categorize the several types of game jams, to critique the model against the current state of the art in the games industry, analyse it as a pre-production process, and finally, define it as a design method. Une game jam est un événement de dévelopement vidéoludique, où des concepteurs de jeu, des programmeurs, des artistes et des musiciens se rassemblent pour créer un jeu dans une période de temps limité. C'est prendre un processus qui prendrait plusieurs années et le compresser dans un ordre de grandeur d'un week-end.
Devenant de plus en plus populaires, il est maintenant important, non seulement pour le monde du game design, mais aussi celui du design, de comprendre les game james et comment elles peuvent être utilisées dans différents contextes et différentes formes.
Le but de ce mémoire de master est de définir et de catégoriser les différents types de game jams, de critiquer le modèle, de regarder à l'état actuel de cet art dans l'industrie, analyser les game jams comme moyen de pré-production et finalement de définir les game jams comme méthode de design.

Contents

List of Figures **4**

List of Tables **7**

1 Introduction **8**

2 Definition **10**
2.1 Chronology of Game Jams 11
2.2 Game Jam Process 16
2.3 List of Game Jams 24
2.4 Design Method 42
2.5 Conclusion 43

3 Narrative Design in Game Jam **44**
3.1 From Genre to Interactive Stories 44
3.2 Narrative in a Mechanic game 47
3.3 Black Whiskers' Narrative Design Process 51
3.4 Author's Narrative Game Jam Entries 54
3.5 Conclusion 56

4 Big Team Game Jam **57**
4.1 Introduction 57
4.2 Big Team Game Jam 58
4.3 Practical Cases 62
4.4 Discussion 67
4.5 Conclusions 69

5 Game Jam as Pre-Production Process **71**
5.1 Post-Jam Measure of Success 74
5.2 Setting up production 76
5.3 Industry Game Jam 78

6 Game Jam as Design Method **82**
6.1 Author's Participation to Game Jams 82
6.2 Exhibition Game Jam 85
6.3 Critics of Game Jam 88
6.4 Future of Game Jams 91

7 Conclusion **92**

8 Credits **94**

List of Figures

1.1 Jammers during the Epic Game Jam #2 8

2.1 Participants of the first Lithuanian Game Jam in August 2002. Courtesy of the Lithuanian Game Jam [27]. 12
2.2 Evolution of the number of participants for the Global Game Jam 14
2.3 Game Jam Process Diagram 15
2.4 View of the Swiss Museum of Games where the Swiss Castle Jam takes place as the Swiss-French Global Game Jam location, in a more setting organizing philosophy. Courtesy of the Swiss Castle Jam [11] 17
2.5 Comparison of the overall satisfaction on a Likert scale between 1 (not at all) and 5 (very much). Results and graph from Pirker J., Voll K. in Group Forming Processes [4]. . . . 19
2.6 A team brainstorming during the Swiss Castle Jam 2016, the Swiss-French Global Game Jam location. 20
2.7 Post-mortem survey of the Ludum Dare 22. Courtesy of Chris Kaitila [6] 21
2.8 What went right survey of the Ludum Dare 22. Courtesy of Chris Kaitila [7] 22
2.9 What went wrong survey of the Ludum Dare 22. Courtesy of Chris Kaitila [7] 23
2.10 The failed prototype of "Fat Cat Rock and Roll" for Ludum Dare 33. 25
2.11 Five different cats in Catpcha, a genetic-mating cat game. . . 27
2.12 A cyborg fights a horde of enemies in STRIDERS, a Cyberpunk Jam entry. 28
2.13 "Heights", the theme of the Cyberpunk Jam. 30
2.14 Over 700 participants ready for the theme revelation at the Nordic Game Jam 2015. 31
2.15 The observation car of the California Zephyr during the 3rd Indie Train Jam 33
2.16 Two jammers during the 0th Swiss Train Jam with Waadt snow mountains in the background. 34

2.17 Jammers during the Epic Game Jam #2 and the subtheme wheel in the background. 35
2.18 A game developer finishing his team's game with kids waiting impatiently during the Kid Game Jam. Courtesy of the Kid Game Jam and David Roulin [14] 37
2.19 Game developers creating levels on Super Mario Maker during the 1st Level Design Game Jam. Courtesy of the Level Design Game Jam [15] 38

3.1 The only room of "I love you no matter what.", a Ludum Dare 31 Jam entry by Team-KwaKwa 45
3.2 A village in the Action-RPG "Power Creep" created during the Global Game Jam 2015 46
3.3 A night in a house in the game "The Stanley Enigma", a Ludum Dare 28 entry by Amos Wenger and Amadine Bru. . 47
3.4 The main character walking in the sewer in the game "The Last Promise" created for the Ludum Dare 27 by the author. 48
3.5 Meow, a hungry cat in the house of a stranger, facing a dog in the video-game "Meow the Great Starvation" 48
3.6 A blue ghost son and his father with a doctor in the Ludum Dare 29 entry "Final Move" 49
3.7 Akeicha on the desert that Earth have become from the "Rise of Akeicha", a Bacon Game Jam entry 50
3.8 The main character climbing a hill with his own thoughts in the game "The Last Promise" a Ludum Dare 27 entry by the author. 51
3.9 Black Whiskers facing his last foe, the masked Big Boss, in the viedo-game "Black Whiskers" created for Game Boy Jam 4 52
3.10 Second page of the author's git commits for "Black Whiskers", the Game Boy Jam 4 entry. 53
3.11 The platformer "Furbrawl", a Game Boy Jam #3 entry . . . 55

4.1 The Swiss-French team of the first I^3 game jam. Courtesy of the I^3 Game Jam [9] 63
4.2 The Swiss Mercenaries Workshop announcement logo. Courtesy of the Swiss Mercenaries [10] 66
4.3 René Bauer (left) and David Javet (right) discussing about Big Team Game Jams during the Ludicious Zurich Game Festival. 68

5.1 A programmer & an artist jamming during Ludum Dare 32 @ Spielhalle Oslo, Basel. 72
5.2 Super Splash Fisticuffs, Project Splash basis, after the Ludum Dare Jam 32. 72

5.3 The setup of Project Splash during Fantasy Basel 75
5.4 Players fighting in the Winter level in Splash Blast Panic, one year after the start of the production. 78
5.5 Tim Schaefer speaking to his employees during the Amnesia Fortnight. Courtesy of Double Fine [97]. 79
5.6 Wasteland Kings, a MOJAM 2 entry by Vlambeer. Courtesy of Vlambeer [64]. 80
5.7 The setup of the "Game Jam" reality show, with some Mountain Dew branding in the background. Courtesy of Robin Arnott and Polygon [90] 81

6.1 The pinata's destruction as theme revelation for the Epic Game Jam #2. Courtesy of the Epic Game Jam [13]. 83
6.2 Prototyping of the Exhibition Game Jam, with the three walls of Motivation, Design and Critics 86
6.3 Game Jam as Design Method 87
6.4 The 0th Indie Game Jam (March 15th to 18th 2002 2.3.2). Courtesy of the Indie Game Jam [38] 89

List of Tables

2.1 Ludum Dare categorization 24
2.2 Game Boy Jam categorization 26
2.3 Bacon Game Jam categorization 26
2.4 PROCJAM categorization 27
2.5 Alt Ctrl Game Jam categorization 28
2.6 Cyberpunk Jam categorization 29
2.7 Nordic Game Jam categorization 31
2.8 Global Game Jam categorization 32
2.9 Indie Train Jam categorization 32
2.10 Swiss Train Jam categorization 33
2.11 Epic Game Jam categorization 34
2.12 Lvl Up Game Jam categorization 35
2.13 Post-Human Game Jam categorization 36
2.14 Indie Game Jam categorization 36
2.15 Kid Game Jam categorization 37
2.16 Level Design Game Jam categorization 38
2.17 NewsGames Hackathon categorization 39
2.18 0hr Game Jam categorization 39
2.19 Game Jams categorization matrix 40
2.20 Game Jams categorization matrix for Category judged in post-jam voting process (5.1) 41

4.1 Categorization of both events 69

6.1 Chronological Author's Participation to Game Jams Table . . 84

Chapter 1

Introduction

Figure 1.1: Jammers during the Epic Game Jam #2

Every young people who are passionate about video-games and have a bit of notion in computer science wants to create a game. They want to create an epic Role-Playing Game or even a Massive Multiplayer Role-Playing Game with the best narration. Obviously, creating such a big scale product with only one person is impossible. Quoting Extra Credits' video "Making Your First Game: Basics - How To Start Your Game Development" [74]

Even if you're amazing and you throw your whole life into cre-

> ating your game, you are not gonna make a God of War or a Final Fantasy. Not even close, and especially not on your first attempt.

Scope is one of the hardest first knowledge learned by game designers and game developers. In game jams, scope is a feature. What is a game jam? It is a game development event where game developers come together to create a game in a short amount of time, normally 48 hours. By design, game jams force participants to scale down their idea to the simplest mechanic.
Game jams are not only events to learn how to scale a game, they are also creative processes, where game developers can try a new way to create games. The author would even define it as "Making game development fun again".
There is a wide range of different types of game jams. The most popular ones, namely the "mainstream" game jams, like the Global Game Jam (2.3.2), the Ludum Dare (2.3.1) or the Nordic Game Jam (2.3.2) are mainly 48-hour event with thousands participants. The game industry has also taken the game jam format its own way, with a more closed and commercial approach, like Amnesia Fortnight (5.3.1), MOJAM (5.3.2), or the failed "Game Jam" reality show attempt (5.3.3). Even a new kind of game jam has started in Switzerland during the year 2015, known as Big Team Game Jam (4), where a big team of more than 10 people tries to create a game in a game jam time-limit format.
From an industry point of view, game jams can be used as a pre-production process to test different ideas. Several game jam entries from the Amnesia Fortnight have then been produced as commercial products. More than a prototype, game jam entries are complete games and new design processes have to be defined to transform a game jam entry into a full commercial product.
Game jams are not only about creating games, it is about the process of creating games in a collaborative and voluntary context. If defined as games, game jams can be playful and gameful, adding a new point of view on how fun can be a game jam.
The author have taken part to the One Game A Month challenge [94] since the end of 2013, using game jams as a main tool to create games of a wide range of different genres. This thesis is about game jams as design method, analyzed through the autoethnographic approach.

Chapter 2

Definition

Definition 1. *A game jam is a development event where participants create a game in a constrained and defined amount of time with some restrictive conditions such as themes.*

The important part of the Definition 1 is the "constrained and defined amount of time". To be more detailed, a game jam has a beginning and a duration, implying an end that are clearly defined before the game jam occurs.

Definition 2. *Game jams are events that are organized to begin at a certain date and time t_1 and to finish at a certain date and time $t_2 > t_1$, known before the event takes place.*

Those definitions (1 & 2) also have consequences. As defined in "participants create a game", the goal of a game jam is to create a game. Due to the restrictions that can be translated as challenges, the goal can be achieved or not, meaning a game jam has a win/lose condition.

Definition 3. *A game jam has challenges and win/lose conditions.*

Using Definition 3, the author defines game jams as games. The paper "Playful Game Jams: Guidelines for Designed Outcomes" [1] defined game

jams as ludic craft, where participation is identified as a constructive form of play. In the same paper, motivations for game jams are intrinsic and participation is voluntary.

Definition 4. *A game jam is a ludic craft game.*

What is the maximum length of a game jam? This is a hard question. Even Gorm Lai refused to answer this question during his interview [98]. This thesis, the author defines the maximum duration of a game jam to one month, as it would be hard to differentiate it to a normal game development process otherwise.

2.0.1 The 0hr Game Jam (2.3.2) case

The 0hr Game Jam (2.3.2) happens during the clock change in Europe. Its duration is 1 hour between 2am Summertime and 2am Wintertime. Its location is in Poznan in the tram circulating around the city. Except the 1 hour restriction and the goal to create a game, there is no other rule.

Definition 5. *The 0hr Game Jam define "some restrictive conditions" from definition 1 as zero. From this case* $\mathsf{some} = \{0, 1, ..., +\infty\}$.

2.1 Chronology of Game Jams

- March 15-18, 2002: 0th Indie Game Jam (2.3.2), 14 game developers created 12 games with the technology theme of "100,000 Guys". The goal was to try to make a game with 100,000 sprites [38].
- April 2002: Ludum Dare Zero, with 18 participants creating 18 games with the theme "Indirect interaction" [20].
- August 24-25 2002: Lithuanian Game Jam, as clone of the Indie Game Jam, with the same goal of creating a game with a lot of sprites. There were 11 programmers (Fig. 2.1) on 6 computers and they ended up with 6 (more or less) completed games and 4 unfinished ones [27].
- February 28-March 3, 2003: 1st Indie Game Jam, 17 game developers created 18 games in 4 days using Zack Simpson et al.'s Shadow Garden, a technology that uses shadow as a interface [39]

Figure 2.1: Participants of the first Lithuanian Game Jam in August 2002. Courtesy of the Lithuanian Game Jam [27].

- May 3-4, 2003 in Kaunas, Lithuania: 2nd Lithuanian Game Jam, 16 programmers created 10 games during these 48 hours with the theme "physics, cars and stuff" [28].
- March 18-March 21, 2004: 2th Indie Game Jam, 21 game developers worked on 17 games over the course of 4 days using Atman Binstock's 2D physics simulator [40].
- December 14, 2005: Darwinia launched on Steam, first third-party game on this platform [78]. A first step in the explosion of the digital video-game market.
- January 27-29, 2006: 1st Nordic Game Jam with 40 participants that created 8 games with the restriction: "create a prototype of a cooperative multiplayer game in one weekend" [82].
- July 11, 2008: App Store launchs [79], like the Steam store, opening a new era of digital video-game market.
- January 30-February 1, 2009: 1st Global Game Jam [16], with the Nordic Game Jam [24] as flagship. The Lithuanian Game Jam 2009 [29], risen from the long inactive period, took place also at the same time as part of the Global Game Jam. With over 1600 participants in 23 countries, and a theme of "As long as we have each other, we will never run out of problems," the GGJ produced 370 games.

- August 20, 2010: Introduction of the Jam category for the Ludum Dare 18, 172 Compo entries and 41 Jam entries with the theme "Enemies as weapons" [21].
- February 15, 2012: 1st MOJAM, a Mojang game jam on humble bundle [61] for charity, a first step for Industry Game Jam. (5.3).
- November 19, 2012: Amnesia Fortnight 2012 [60] on humble bundle by Double Fine Productions. A first step in the Commercial Game Jam category (Definition 23).
- January 18-20, 2013: 8th Nordic Game Jam, no more as flagship for the Global Game Jam, with 470 participants. In his interview [98], Gorm Lai explained this separation by the will of The Nordic Game Jam to become more independent from the overall big structure of the Global Game Jam. The theme was "Grotesque" [25].
- January 25-27, 2013: 5th Global Game Jam, with 319 sites, 63 countries, 16,705 registered participants, 3248 games. The theme was the sound of a heartbeat.
- February 20, 2013: 2nd MOJAM, a Mojang game jam on Humble Bundle [61] for charity, with other indie game studios: Grapefrukt, Ludosity, Oxeye Game Studio, Vlambeer, and Wolfire Games.
- Feb 6, 2014: Amnesia Fortnight 2014 [62] on Humble Bundle by Double Fine Productions.
- March 13-15, 2014: 1st Indie Train Jam, 60 developers created 18 games with the theme "Disconnected" [22]. The event was widely made famous from the movie "Gameloading - Rise of the Indies"that featured it [23].

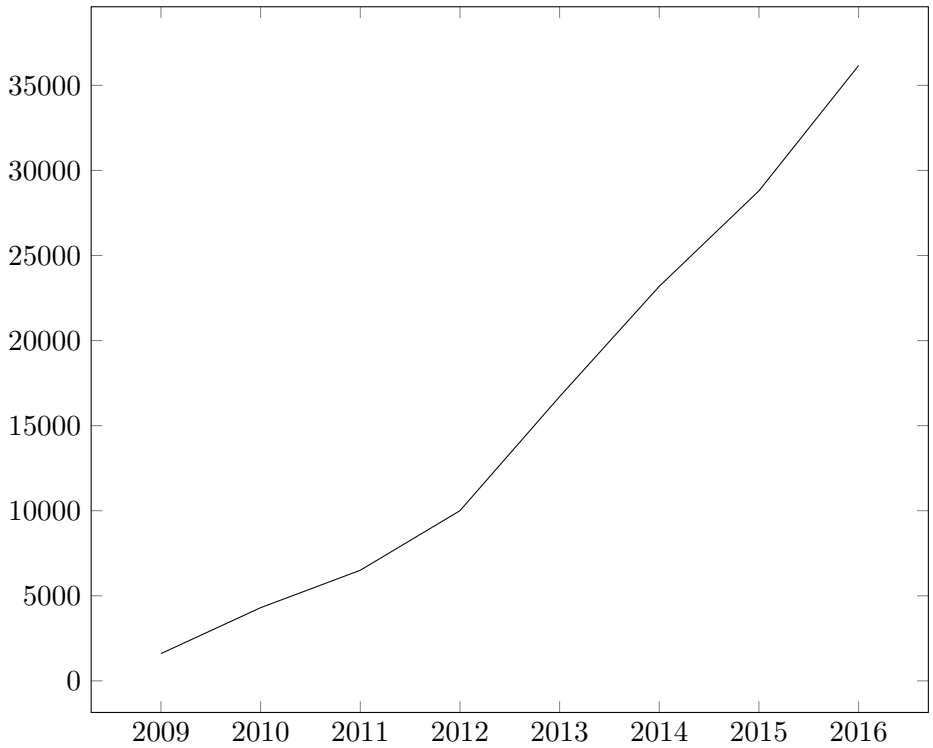
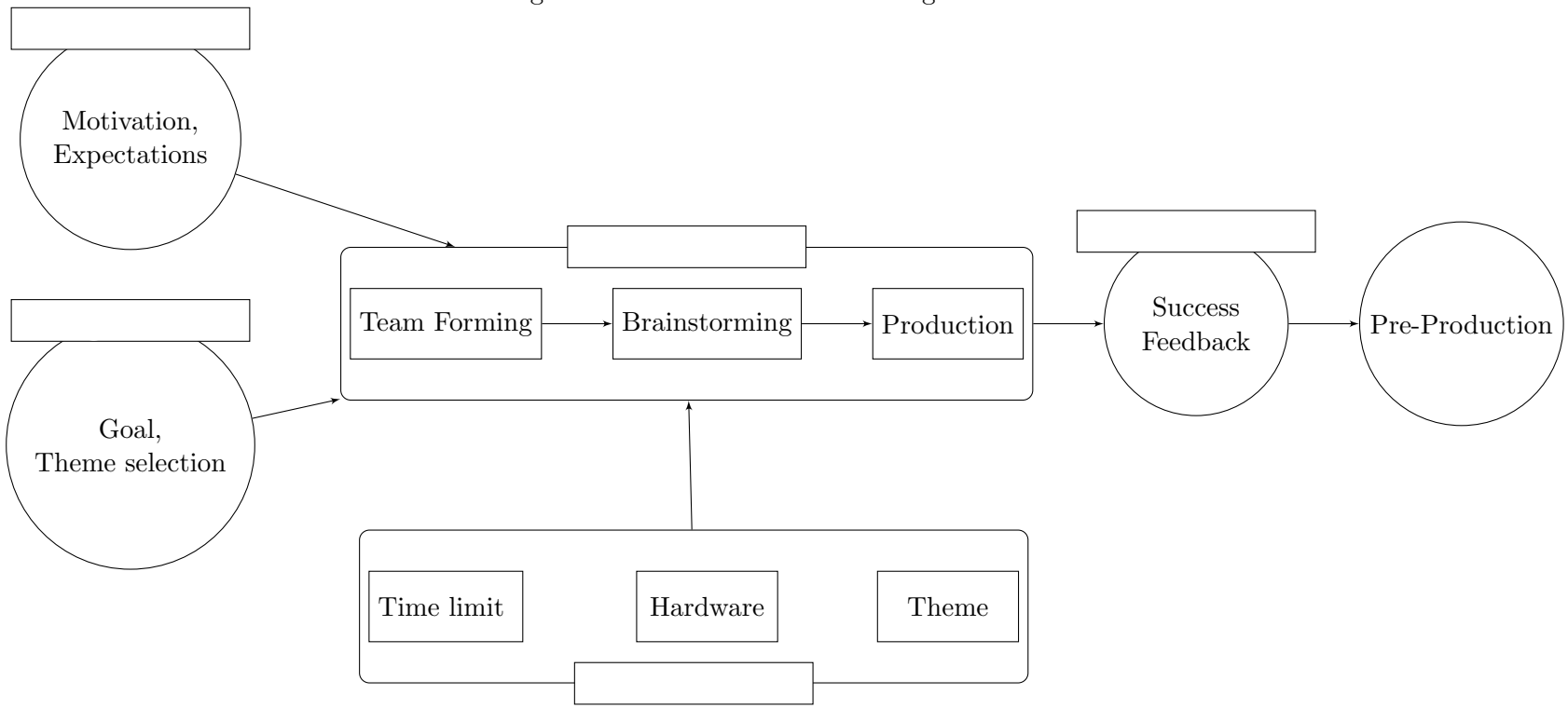


Figure 2.2: Evolution of the number of participants for the Global Game Jam

Figure 2.3: Game Jam Process Diagram



2.2 Game Jam Process

Like defined in Definition 2, the time-limit format is the basis of the game jam. Of course, game jam organizers select a certain date for the game jam with several restrictions and goal, and, often, with a certain theme. Participants come with a intrinsic motivation (Definition 4), but also with certain expectations.
During the event, participants form team, then brainstorm to get a game idea, and finally produce their game. At the end of the event, people can play the produced game and give their opinion about it. Depending on the post-jam success, participants can decide to continue the development of the game outside of the game jam event (5).
More than an event, a game jam is an process (Table 2.3) with different varieties around the organizing philosophy [3]. Those variations can be measured through the lens of play on the playful versus gameful continuum [1].

2.2.1 Playfulness

From Definition 4 and the paper "Playful game jams: Guidelines for Designed Outcomes" ([1]), game jam are defined as Ludic Craft. For the authors of the paper, a game jam is a game. From this basis, it is possible to apply the lens of play on the playful versus gameful continuum:

- Playful is unstructured, spontaneous, open-ended; in Caillois' term [41]: paidia.
- Gameful is structured, rules based; in Caillois' term [41]: ludus.

It is important to understand that every decisions concerning the game jam event is going to be seen on the lens of play. The author, using the framework at disposal from [1], defines if each jam from 2.3 has a playful or gameful experience and outcome.

2.2.2 Organizing Philosophies

The first question that comes in mind when trying to organize a game jam is "why?". What kind of experiences will the participants have? Those questions are answered by choosing an organizing philosophy. The organizing philosophy defines the game jam experience, the pitch of the event expresses the organizing philosophy. There are six different kinds of organizing philosophies [3]:

- Regional affiliation: around a specific region (Nordic game jam, Global game jam, etc...)

Figure 2.4: View of the Swiss Museum of Games where the Swiss Castle Jam takes place as the Swiss-French Global Game Jam location, in a more setting organizing philosophy. Courtesy of the Swiss Castle Jam [11]

- Setting: draw intrigue from unusual setting or exceptionally beautiful (Swiss Castle Jam (Fig. 2.4), Indie Train Jam (2.3.2)).
- Technology: particular tool or technology limitations (Game Boy Jam (2.3.1), PROCJAM (2.3.1))
- Career advancement: clear monetary goal or career recognition.
- Social/Charitable topics: specific niches in certain non-profit communities.
- Challenge: Arbitrary challenges to create "fun" motivation. Game Design as motivation.

Those categories targets different game developers' audience. For example, regional affiliated game jams targets developers from the region.

2.2.3 Theme Selection

Many game jams have communities, that support the game jam outside of the scope of the event [1]. In certain game jams (like Ludum Dare 2.3.1 or Bacon Game Jam2.3.1), the theme selection is done by the community through a voting process.
This process can be more complex for bigger event, like for example the Global Game Jam. For the Global Game Jam 2012, Annakaisa Kultima says in [56]:

> There were 95 distinguishable ideas for the GGJ 2012 theme in total. Out of 95 suggested ideas, 44 were commented on by others. 17 ideas received support from another committee member and 27 ideas were questioned or refuted by someone.

For Annakaisa Kultima, the theme has different aspect. It is an Event Design element, that sets the tone of the game jam experience. It is the start of the event. It is a Creative Stimulus, a tool for brainstorming (2.2.5). It is a Design Constraint and how she explained it in the context of the Global Game Jam committee [56]:

> [...]the theme was also framed by the committee members as a problem to solve, a challenge or explicitly framed idea for the participants to engage with.

Finally, the theme is also an Experiment and an Example: "An experimental theme was seen as an example that could encourage participants to be more daring". It is confirmed by Gorm Lai in his interview [98], that praises the Global Game Jam as a creative and experimental event.
For the author, the theme is the first step of the jam, a focus point. From all the game jams he participated in, the author remembers all the themes. They are a very important part of a game jam.

2.2.4 Team formation

In some game jams, people can form team during the event. On the playful versus gameful continuum, those game jams are more playful than game jams where people have to register as a team, as quoted from [1]:

> This enables the participants to find a cohesive team with a good balance of personality types and skill sets. However, this may limit the social outcomes (e.g. networking) and increases the barrier to entry.

For example, the Global Game Jam (2.3.2) answers to the question "Do I need to come to GGJ with a team already formed?" is:

> Do not come to the Jam with a team. Everyone will have some time to think and pitch an idea. Collaborate with new friends or peers you admire.

The paper "Group Forming Processes - Experiences and Best Practice from Different Game Jams" ([4]) defines three categories of team formation during the event:

- Unsupervised Group Forming ([4] Section 2.1)
- Supervised Group Forming (including Skillset) ([4] Section 2.2)

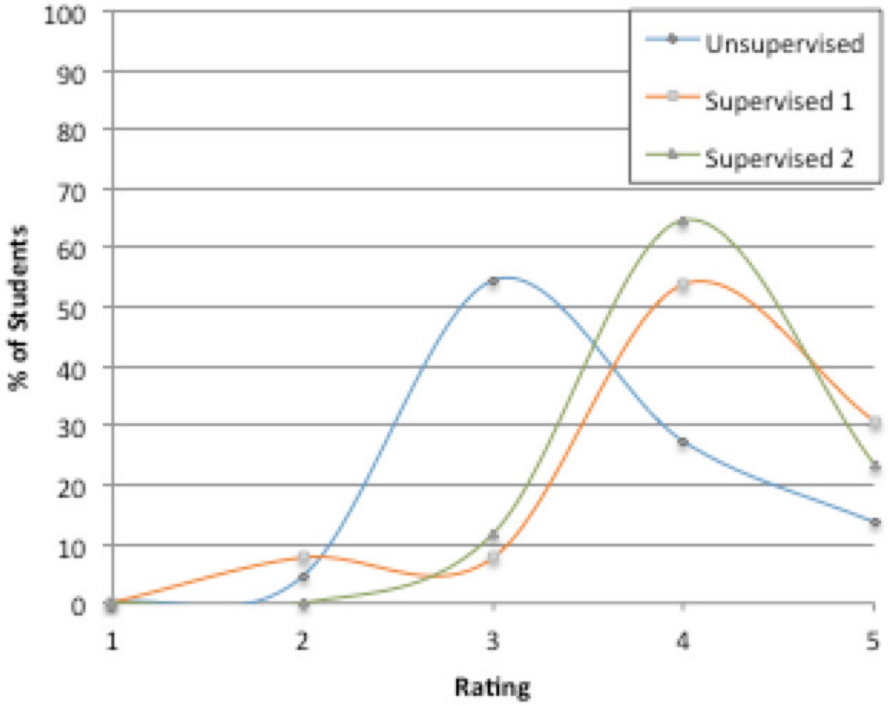


Figure 2.5: Comparison of the overall satisfaction on a Likert scale between 1 (not at all) and 5 (very much). Results and graph from Pirker J., Voll K. in Group Forming Processes [4].

- Supervised Group Forming (including Skillset and Tools/Engine) ([4] Section 2.3)

According to the paper, satisfaction increases with more supervised group forming (Figure 2.5).

2.2.5 Brainstorming

Definition 6. *The brainstorming phase of a game jam allows team to define the main mechanic, the specific genre of the game jam entry, and the narrative of the game.*

After the Team Formation phase, team comes together to brainstorm the game idea. Like defined in Definition 6, Brainstorming can be seen as the Pre-Production process of a game jam. Not defined as a Design Method

Figure 2.6: A team brainstorming during the Swiss Castle Jam 2016, the Swiss-French Global Game Jam location.

in [43], brainstorming could be used as a planning, scoping, and definition phase.

2.2.6 Post-Jam

There are some game jams that organize after the event a voting process. For example, the Ludum Dare (2.3.1) voting process lasts three weeks. Participants always vote using several categories (Table 2.20). From the paper "Playful Game Jams" [1], a breadth of incompatible awards brings more playfulness to a game jam, though objective awards brings more gamefulness.
A common practice after a game jam is to write a post-mortem with the typical "What went right" (Figure 2.8) and "what went wrong" (Figure 2.9) structure. Those post-mortem allow the developers to have an objective point of view on their own experience and share it to the world.
The author talks about the post-jam process in the chapter about Game Jam as Pre-Production process (5.1).

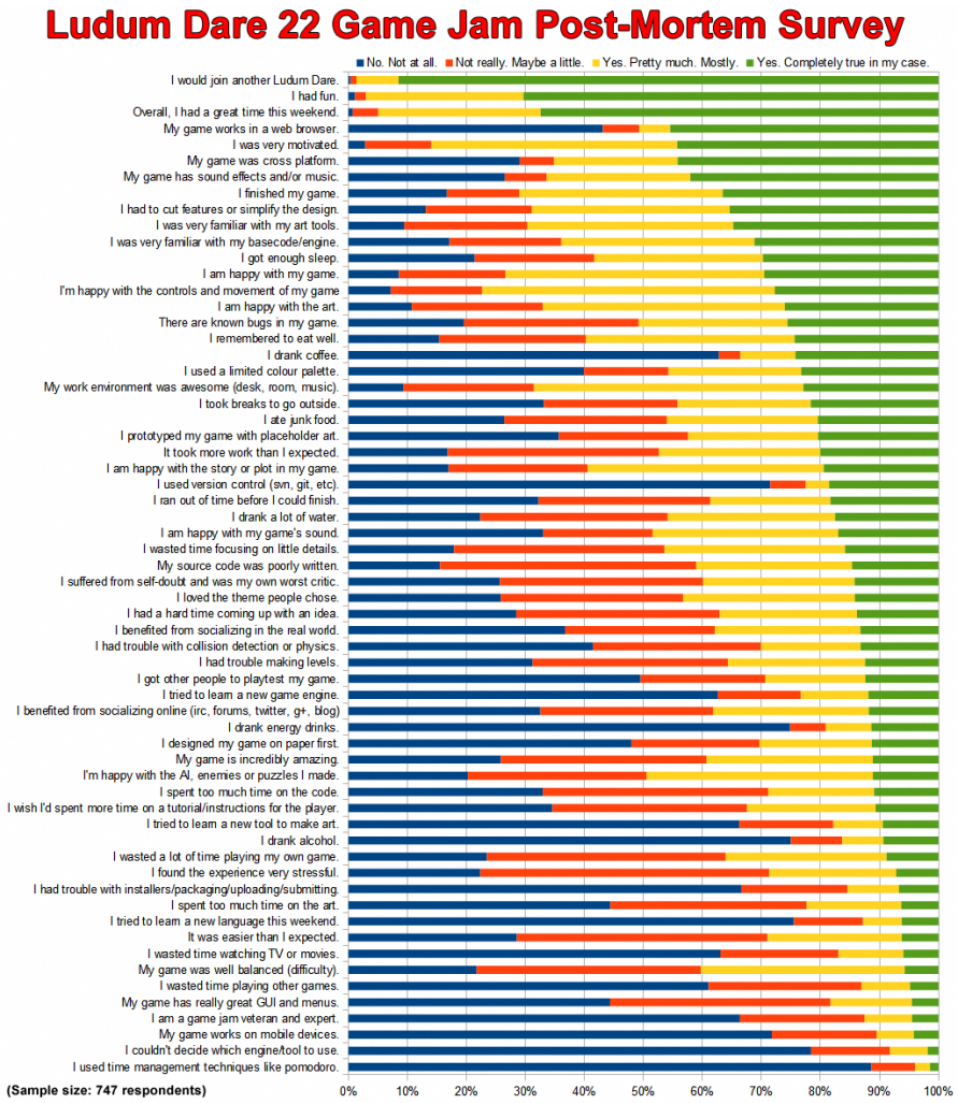


Figure 2.7: Post-mortem survey of the Ludum Dare 22. Courtesy of Chris Kaitila [6]

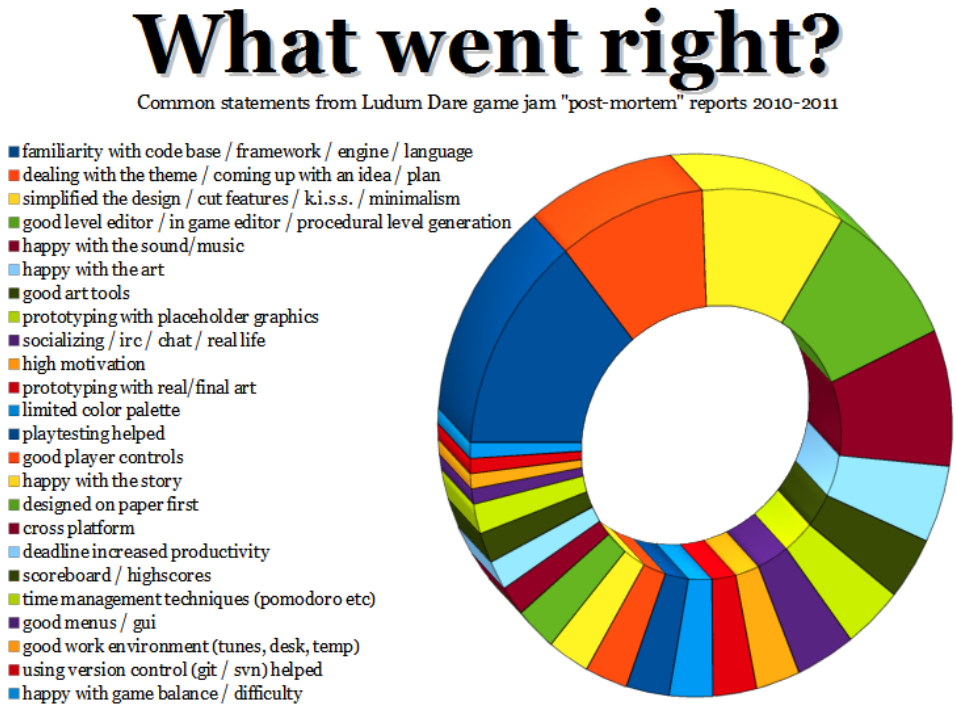


Figure 2.8: What went right survey of the Ludum Dare 22. Courtesy of Chris Kaitila [7]

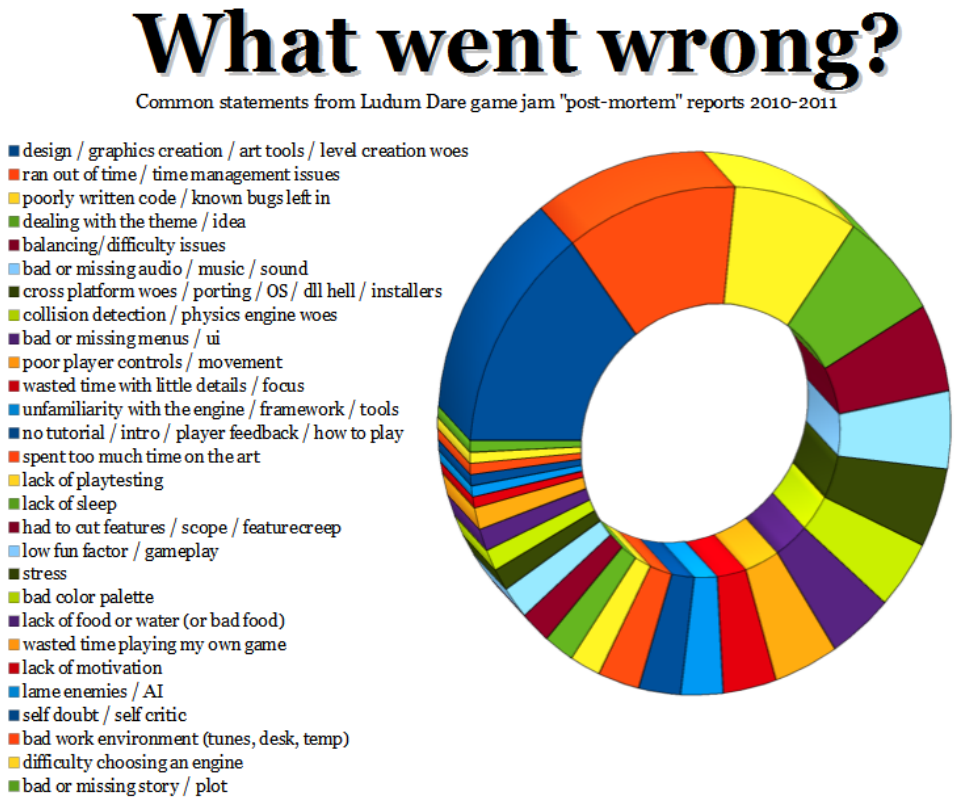


Figure 2.9: What went wrong survey of the Ludum Dare 22. Courtesy of Chris Kaitila [7]

2.3 List of Game Jams

The author lists in this section the game jams that are present in this thesis. A lot of other game jams occur every year, but it would be impossible to list them all here.

2.3.1 Online Game Jams

Online game jams are generally focused on a website or webpage. People can unofficially create locations around the world and can also work from home.

Ludum Dare

The Ludum Dare [17] is a tri-annual online game jam that started in 2003. It is divided in two categories:

- Compo
- Jam

The Compo is the original concept of Ludum Dare. It is a solo competition, where the participant create a game from scratch, meaning that he codes, he draws, and makes his own musics and sounds in 48 hours
The Jam is a 2010 addition to the event, where the only restriction is the theme. The game can be made solo or in team. The Jam duration is 72 hours.

Table 2.1: Ludum Dare categorization

	Compo	Jam
Restrictions	Theme No Team Content Creation	Theme
Duration	48 hours	72 hours
Organizing Philosophy	Challenge	idem
Group Forming	-	Unsupervised
Experience	Gameful	idem
Outcome	Playful	idem
Judged categories	Overall, Innovation, Fun, Theme, Graphics, Audio Humor, and Mood from 1-5	idem

The author participated in several Ludum Dare, mostly the Jam category with a team.
The author tried the Ludum Dare Compo, but failed to deliver a result [75], explained by a motivation issue due to lack of group pressure.

Here is the list of games created by the author at Ludum Dare:

- The Last Promise [19]
- Final Move [20]
- I love you no matter what [8]
- Fat Cat Rock and Roll (Fig. 2.10) [75]

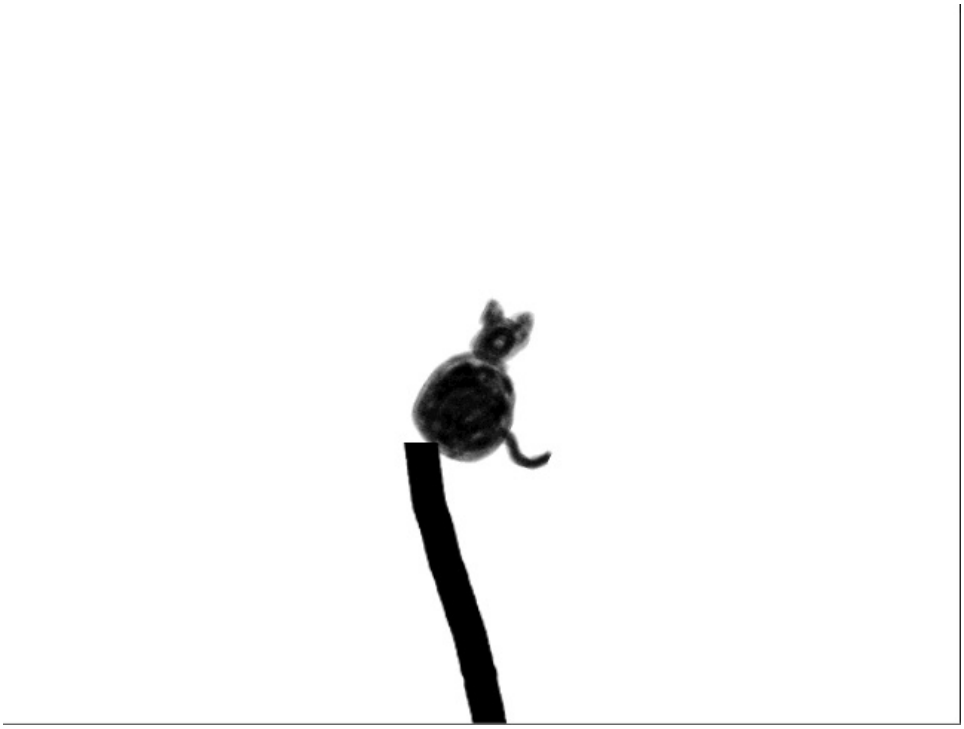


Figure 2.10: The failed prototype of "Fat Cat Rock and Roll" for Ludum Dare 33.

Game Boy Jam

The Game Boy Jam [87] is a recent annual online game jam. Its duration is 10 days (a long jam) and has the only restriction that the game has to feel like a Game Boy game. It means 160x144 pixels and 4 colors by palette. There is no theme, except the Game Boy feeling.
The longer duration allows for developers used to the 48-hour game jam format to scale up their game jam entry. For example, the author participated in two Game Boy Jams, trying to bring a narrative approach to its games, for example the game "Black Whiskers" (3.4.8) has three levels, with three end bosses that have dialogues telling a longer story.

Table 2.2: Game Boy Jam categorization

	GBJAM
Restrictions	Game Boy Color palette, resolution, input.
Duration	10 days
Organizing Philosophy	Technology
Group Forming	Unsupervised
Experience	Gameful
Outcome	Playful
Judged categories	Overall, Graphics, Sound, Gameplay, GameBoy, Overall user score from 1 to 5

Bacon Game Jam

The Bacon Game Jam [88], originally Reddit Gamedev Jam, is a tri-annual online game jam. The theme is voted on a reddit comment page by reddit users. Its duration is 48 hours.

Table 2.3: Bacon Game Jam categorization

	Bacon GJ
Restrictions	Theme.
Duration	48 hours
Organizing Philosophy	Challenge
Group Forming	Unsupervised
Experience	Gameful
Outcome	Playful
Judged categories	Gameplay, Graphics, Audio, Innovation, Story, Technical, Humor, and Controls from 1-10

The first author's game jam was the Bacon Game Jam 6. Since then, he participated to several of them, always trying new concepts. Here are the list of its games:

- Trials [1], a ray-puzzle game for Bacon Game Jam 6.
- Meow: the Great Starvation [21], an infiltration game for Bacon Game Jam 7.
- Millions of Aliens of Doom [6], a new take on shoot them up where the player controls the final boss and throw projectiles.
- Rise of Akeicha [11], a narrative point-and-click adventure set in a post-apocalyptic world where people live underground.

PROCJAM

PROCJAM or the Procedural Generation Jam [89] is an annual game jam/hackathon where participants during one week *make something that makes something* [89].
The author participated in the 2015 edition and created Catpcha [14], a genetic-mating cat game (Fig. 2.11).

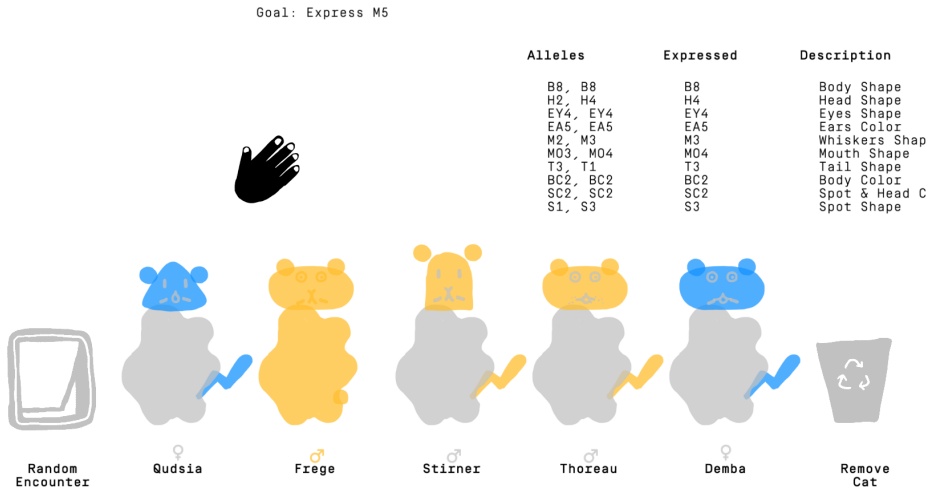


Figure 2.11: Five different cats in Catpcha, a genetic-mating cat game.

Table 2.4: PROCJAM categorization

	PROCJAM
Restrictions	Procedural
Duration	10 days
Organizing Philosophy	Technology
Group Forming	Unsupervised
Experience	Gameful
Outcome	Gameful

Alt Ctrl Game Jam

The Alternative Controller Game Jam [18] is an online 10-day, where participants create a game using alternative controllers. The author did not yet participate in the Alt Ctrl Game Jam.

Table 2.5: Alt Ctrl Game Jam categorization

	Alt Ctrl Game Jam
Restrictions	Theme, Hardware
Duration	10 days
Organizing Philosophy	Technology
Group Forming	Unsupervised
Experience	Gameful
Outcome	Playful

Cyberpunk Jam

The Cyberpunk Jam [80] was an online 12-day game jam, where participants had to create a game in a cyberpunk universe with the theme "Height" (Fig. 2.13).
The author participated during the only edition in 2014 and created STRIDERS [2], a runner shooter (Fig. 2.12).

Figure 2.12: A cyborg fights a horde of enemies in STRIDERS, a Cyberpunk Jam entry.

Table 2.6: Cyberpunk Jam categorization

	Cyberpunk Jam
Restrictions	Theme
Duration	12 days
Organizing Philosophy	Challenge
Group Forming	Unsupervised
Experience	Gameful
Outcome	Playful

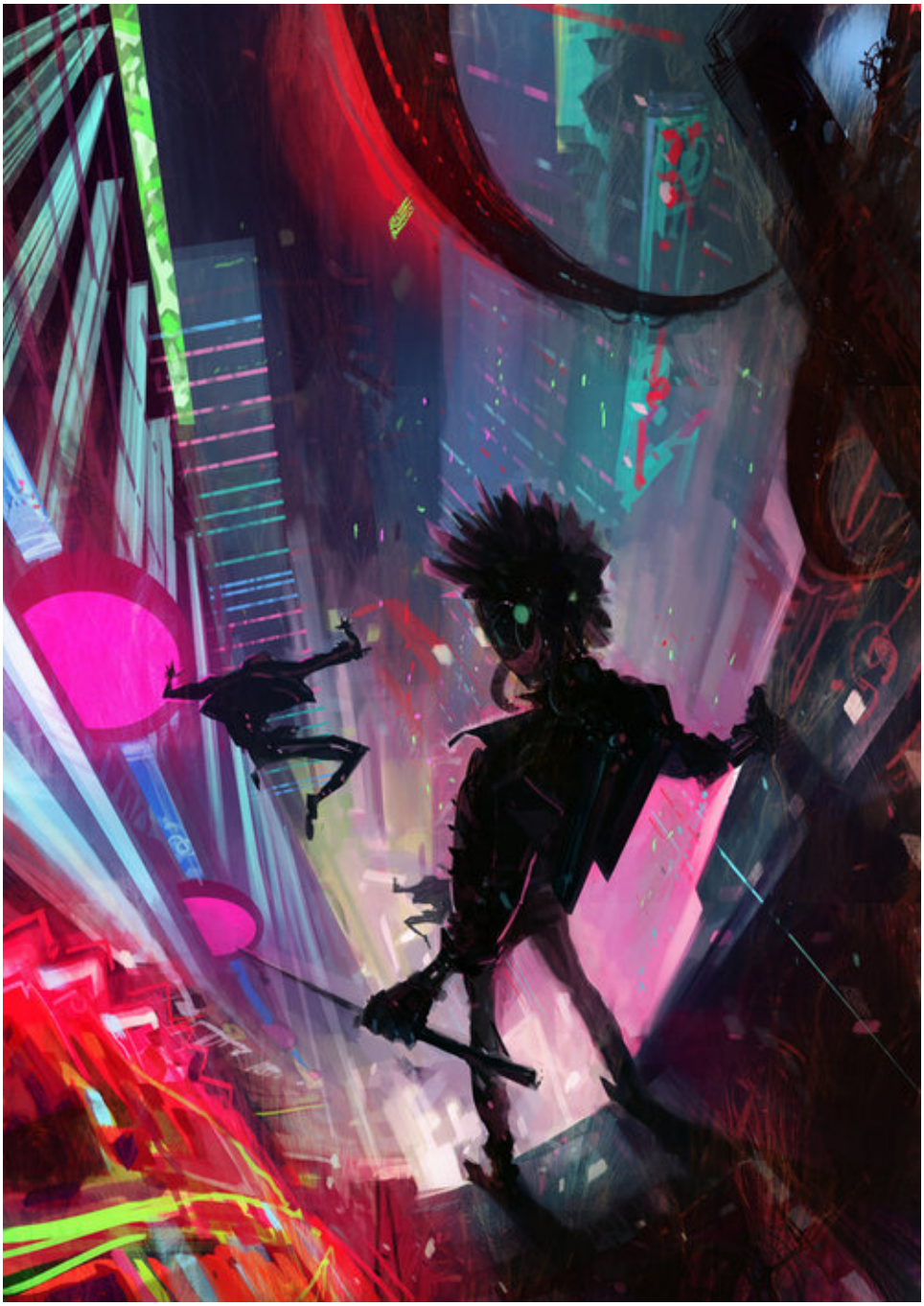


Figure 2.13: "Heights", the theme of the Cyberpunk Jam.

2.3.2 Local Game Jams

Local Game Jams have official locations where participants come together.

Nordic Game Jam

Nordic Game Jam [24] is the biggest annual local game jam in the world. The 2015 edition gathered 700 participants for 48 hours in Copenhagen (Fig. 2.14). People around the world comes to jam together.
In his interview [98], Gorm Lai, one of the founders of the Nordic Game Jam, explained how they took inspirations from the Indie Game Jam (2.3.2) and how the Nordic Game Jam helped create game studios around the Nordic game development community.
The author participated in the 2015 edition, creating some mini-games packed into "Wash Machine Simulator" [9].

Figure 2.14: Over 700 participants ready for the theme revelation at the Nordic Game Jam 2015.

Table 2.7: Nordic Game Jam categorization

	Nordic GJ
Restrictions	Theme.
Duration	48 hours
Organizing Philosophy	Regional affiliation
Group Forming	Supervised
Experience	Gameful
Outcome	Playful

Global Game Jam

Inspired by Nordic Game Jam, Global Game Jam [16] is an annual local game jam all around the world. Locations are set up in a lot of city around the world. The biggest location in 2016 was in Giza, Egypt.

The Global Game Jam started with the Nordic Game Jam as flagship in 2009. After several years, the Nordic Game Jam separated itself from the Global Game Jam to be more independent from the structure of the Global Game Jam. The Global Game Jam did not suffer from this separation and continued its growth, always breaking its own record of participation.

Table 2.8: Global Game Jam categorization

	GGJ
Restrictions	Theme.
Duration	48 hours
Organizing Philosophy	Regional affiliation
Group Forming	Unsupervised
Experience	Gameful
Outcome	Gameful

Indie Train Jam

The Indie Train Jam [22] is an annual game jam taking place in a train between Chicago and San Fransisco before the Game Developer's Conference. Its duration is the same as the trip, which is 52 hours. In his interview [98], Gorm Lai praises the social benefits of such a closed setting. Compare to other local game jams, participants can not leave the jam location. From the author experience, the linear setting also helps free agents moving easily from one group to another, compared to a two dimensional setting that is often seen in local game jam in term of spatial distribution.

Table 2.9: Indie Train Jam categorization

	Train Jam
Restrictions	Theme.
Duration	52 hours
Organizing Philosophy	Setting
Group Forming	Unsupervised
Experience	Playful
Outcome	Playful

Swiss Train Jam

The Swiss Train Jam [95] is Indie Train Jam clone, adapted to Switzerland. Traveling 9 hours around Switzerland, changing train every 20 minutes to 2 hours, the 0th edition occurred February the 14th 2016, organized by the author.

Figure 2.15: The observation car of the California Zephyr during the 3rd Indie Train Jam

Table 2.10: Swiss Train Jam categorization

	Swiss Train Jam
Restrictions	Theme.
Duration	9 hours
Organizing Philosophy	Setting
Group Forming	Supervised
Experience	Playful
Outcome	Playful

Epic Game Jam

The Epic Game Jam [13] is an annual local game jam organized mainly in Neuchatel and is strongly tied to the Swiss-French game developer community. Its duration is 48 hours and added to the main theme comes during the duration of the week-end other sub-themes. The judges use a huge amount of different badges to award in a playful way [1], a wide range of incompatible awards. The author participated in in the two editions of the Epic Game Jam.

Figure 2.16: Two jammers during the 0th Swiss Train Jam with Waadt snow mountains in the background.

Table 2.11: Epic Game Jam categorization

	Epic Train Jam
Restrictions	Theme.
Duration	48 hours
Organizing Philosophy	Challenge
Group Forming	Unsupervised
Experience	Gameful
Outcome	Playful

Lvl Up Game Jam

The Lvl Up Game Jam [12] is a two-day local game jam organized at the Engineering School of Fribourg. Teams are divided in two categories:

- amateur
- professionals

Both of those categories have three different themes and they have to choose at least one. At the end of the game jam, an award ceremony celebrates the best games in the different judged categories (Table 2.12).

Figure 2.17: Jammers during the Epic Game Jam #2 and the subtheme wheel in the background.

Table 2.12: Lvl Up Game Jam categorization

	Lvl Up Game Jam
Restrictions	Themes
Duration	48 hours
Organizing Philosophy	Challenge
Group Forming	Unsupervised
Experience	Gameful
Outcome	Playful
Judged categories	Originality, Artistic Quality, Technical Quality, Gameplay

Post-Human Game Jam

Post-Human Game Jam [76] was a 24 hours (from Saturday noon to Sunday noon) that happened during the Symposium PI in Monthey, Wallis. The theme was large about the future of human being and humanity with several diversifiers.

Table 2.13: Post-Human Game Jam categorization

	Epic Train Jam
Restrictions	Theme.
Duration	24 hours
Organizing Philosophy	Challenge
Group Forming	Unsupervised
Experience	Gameful
Outcome	Playful

Indie Game Jam

Organized by Chris Hecker and Jonathan Blow, it was the first game jam in 2002. The author gives more detail in his chronology (2.1).

Table 2.14: Indie Game Jam categorization

	Indie Train Jam
Restrictions	Technology.
Duration	4 days
Organizing Philosophy	Challenge
Group Forming	Unsupervised
Experience	Gameful
Outcome	Playful

Kid Game Jam

The Kid Game Jam [14] is a 48-hour game jam, where game developers are creating a game with children having decision design control. Game developers translate the children's vision into the computer.

Figure 2.18: A game developer finishing his team's game with kids waiting impatiently during the Kid Game Jam. Courtesy of the Kid Game Jam and David Roulin [14]

Table 2.15: Kid Game Jam categorization

	Kid Train Jam
Restrictions	Children.
Duration	48 hours
Organizing Philosophy	Challenge
Group Forming	Supervised
Experience	Playful
Outcome	Playful

Level Design Game Jam

Organized by Sandro Dall'Agio in 2015, the goal of the Level Design Game Jam [15] was for participants to only focused on the Level Design. The first edition was done using the game Super Mario Maker.

Figure 2.19: Game developers creating levels on Super Mario Maker during the 1st Level Design Game Jam. Courtesy of the Level Design Game Jam [15]

Table 2.16: Level Design Game Jam categorization

	Level Design Train Jam
Restrictions	Level Design.
Duration	11 hours
Organizing Philosophy	Challenge
Group Forming	Unsupervised
Experience	Gameful
Outcome	Gameful

NewsGames Hackathon

Even if the title could be confusing, the NewsGames Hackathon [77] was a game jam, where participants had to create games about "news" in a large context. Its duration was an afternoon.

Table 2.17: NewsGames Hackathon categorization

	NewsGames Hackathon
Restrictions	Theme
Duration	4 hours
Organizing Philosophy	Challenge
Group Forming	Unsupervised
Experience	Gameful
Outcome	Gameful

0hr Game Jam

The 0hr Game Jam [36] is a game jam that occurs in Poznan and online during the clock change in Europe and in America. Its duration is the 1 hour between 2am Summer Time and 2am Winter Time.

Table 2.18: 0hr Game Jam categorization

	0hr Game Jam
Restrictions	-
Duration	1 hour
Organizing Philosophy	Challenge
Group Forming	Unsupervised
Experience	Playful
Outcome	Playful

Table 2.19: Game Jams categorization matrix

Name	Location	Durations	Restrictions	Organizing Philosophies	Group Forming	Experience	Outcome
Ludum Dare Compo	Online	48 hours	Theme	Challenge	-	Gameful	Playful
Ludum Dare Jam	Online	72 hours	Theme	Challenge	Unsupervised	Gameful	Playful
Bacon Game Jam	Online	48 hours	Theme	Challenge	Unsupervised	Gameful	Playful
Game Boy Game Jam	Online	10 days	Game Boy feeling	Technology	Unsupervised	Gameful	Playful
PROCJAM	Online	10 days	Procedural	Technology	Unsupervised	Gameful	Gameful
Alt Ctrl Game Jam	Online	10 days	Theme Hardware	Technology	Unsupervised	Playful	Playful
Nordic Game Jam	Local	48 hours	Theme	Regional	Supervised	Gameful	Playful
Global Game Jam	Local	48 hours	Theme	Regional	Unsupervised	Gameful	Gameful
Indie Train Jam	Local	52 hours	Theme	Setting	Unsupervised	Playful	Playful
Epic Game Jam	Local	48 hours	Theme	Regional	Unsupervised	Gameful	Playful
Lvl Up Game Jam	Local	48 hours	Themes	Challenge	Unsupervised	Gameful	Playful
Post-Human Jam	Local	24 hours	Theme	Challenge	Unsupervised	Gameful	Playful
Indie Game Jam	Local	4 days	Technology	Challenge	Unsupervised	Gameful	Playful
Kid Game Jam	Local	48 hours	Children	Challenge	Supervised	Playful	Playful
Level Design Game Jam	Local	11 hours	Level Design	Challenge	Unsupervised	Gameful	Gameful
NewsGames Hackathon	Local	4 hours	Theme	Challenge	Unsupervised	Gameful	Gameful
0hr Game Jam	Local	1 hour	-	Challenge	Unsupervised	Playful	Playful

Table 2.20: Game Jams categorization matrix for Category judged in post-jam voting process (5.1)

Name	**Judge categories**
Ludum Dare Compo and Jam	Overall, Innovation, Fun, Theme, Graphics, Audio, Humor, and Mood from 1-5
Bacon Game Jam	Gameplay, Graphics, Audio, Innovation, Story, Technical, Humor, and Controls from 1-10
Game Boy Jam	Overall, Graphics, Sound, Gameplay, Game Boy, Overall user score from 1 to 5
Cyberpunk Jam	Aesthetics, Relevance, Overall Synergy, Polish from 1 to 5
Lvl Up Game Jam	Originality, Artistic Quality, Technical Quality, Gameplay

2.4 Design Method

Design methods are process format helping to support the design process. John Chris Jones defines design methods on his website [71] as:

> Techniques which enable people to design something, to go beyond their first ideas, to test their designs in use or simulated use, to collaborate in creative activity, to lead design groups and to teach and to learn designing. A method can be anything one does while designing: sketching alternative designs 'on the back of an envelope', calculating what are assumed to be the main parameters, formal brainstorming (and classification of the result), taking a rest, issuing a questionnaire, evaluating preliminary designs in 'affirmative groups' and, most importantly, observing and experiencing for oneself the use of existing or new designs (in real life or in simulations)... A design method is any action whatever that the designers may decide is appropriate.

From this quote, it is important to directly define:

Definition 7. *Design methods are actions, that support the design process.*

John Chris Jones defines them as techniques, someone one does. For the author, they are clearly separated of design tools:

Definition 8. *Design methods are objects, that support the design process.*

Design methods and tools are part of a design process. Every designer has his own design process. John Chris Jones defines his design process like this [44]

1. Divergence, the act of extending the boundary of a design situation so as to have a large enough, and fruitful enough, search space in which to seek a solution.

2. Transformation, pattern-making, fun, flashes of insight, changes of set.

3. Convergence, the problem has been defined, the variables have been identified and the objectives have been agreed. The designer's aim ...[is to] reduce the secondary uncertainties progressively until only one of many possible alternative designs is left...

An other perspective on design process is found in "Universal Methods of Design" [43]. Each of the hundred methods presented are divided in the five design phases:

1. Planning, Scoping, Definition
2. Exploration, Synthesis, Design Exploration
3. Concept Generation and Early Prototype Generation
4. Evaluation, Refinement, Production
5. Launch and Monitor

A design method produce a result through its action. A design method has a goal, not just to help the designers, but to make them advance in the design process.

2.4.1 Prototyping

Prototyping is defined as a design method in "Universal Methods of Design" [43]:

> Prototyping is the tangible creation of artifacts at various level of resolution, for development and testing of ideas within design teams and with clients and users.

A rapid prototyping can, but is not automatically a game jam. The event that is a game jam is a subset of a rapid prototyping session. The difference between a prototype product and a game jam entry is discussed more widely in the chapter 5.

2.5 Conclusion

All those definitions help the author base his thesis on an understanding common ground. Defining game jam is as hard as defining what a game is. The current definition focuses on the time-limit format that is mandatory for all game jams. Design method also had to be defined clearly before going on with the rest of the thesis.
Looking back at the history of game jams is an important step for this thesis, to see where the movement emerged and how it became so successful and democratized.
Game jams have their own process, divided in time before, during, and after the event. The author listed and categorized using the location of the game jam, the duration, the organizing philosophies, the team formation process, the playfulness of the experience and outcome, the restrictions, and the judged categories.

Chapter 3

Narrative Design in Game Jam

Game jams are not known to create compelling narrative. One of the main reasons is that the time constraint of game jam does not allow for depth in the game creation, an argument from Jonathan Blow in his blog post about the Depth Jam[51].
When scaling down the game development process from several years to several days, the different parts that constitute the process are also scaled down. It is the same with the narrative design of a game.
There is a constant struggle between narration and gameplay in the game development process. Some game jams' participants try to create a narrative game and this chapter is a framework to create compelling narrative in short-term game jam.

3.1 From Genre to Interactive Stories

The brainstorming process of a game jam often revolves around the definition of the main mechanic or the specific genre of the game jam entry before going into the narrative. This section gives tools for several narrative genres.

3.1.1 Point and Click Adventure

Point and Click Adventure is a video-game genre that peaked during the nineties with games like Monkey Islands and that came back recently with games like The Walking Dead [1] or Life is Strange [2]. This genre is known for very compelling interactive stories, but also for very frustrating and complex puzzles.
A point and click adventure is several hours long and it is impossible to create a game of this scale in the span of a weekend. Fortunately, point and click adventure are divided in several rooms. A manageable goal in a

weekend is to create one to two compelling rooms. "I love you no matter what" (Figure 3.1 3.4.5) , a game created by the author during the Ludum Dare 31 is a perfect example of this one-room-based point and click. The theme of the game jam was "Entire game in one screen", making it more than a recommendation.
Then, if the game jam was set up as a pre-production process, the post-jam process of feedback can settle if the narrative tools were well used and the story compelling enough on the short-term mechanic motivation.

Figure 3.1: The only room of "I love you no matter what.", a Ludum Dare 31 Jam entry by Team-KwaKwa

3.1.2 Role-Playing Game

Role-Playing Game (or RPG) is a genre that inspired a lot of young game designers. The trend for beginner game developers is to create a JRPG or a MMORPG, because they were games that connected deeply with them. The problem is that any types of RPG are extremely complex games to create, even for an experienced game developer. The challenge is the gigantic amount of content that is needed.
In the context, it is impossible to create a complete RPG. Like the Point and Click Adventure genre (3.1.1), the idea is to scale down to one dungeon (for combat-based RPG) or one village (for a more narrative-based game), depending on the initial motivation of the game developer. "Power Creep" (Fig. 3.2 3.4.6), a game created by the author is a concrete example of a little Action-RPG.

Figure 3.2: A village in the Action-RPG "Power Creep" created during the Global Game Jam 2015

3.1.3 Text-Based Game

Creating a text-based game jam entry poses some problems not only in the creation process, but also for reception. One of the first problems is the language barrier. Even if English is widely spread, not everyone can read or write in proper English.
More than that Text-based games are a niche. Like seen in the Pre-Production chapter 5.1, reaction to text-based games can be worse and more superficial than expected compared to the work done. As Text-based games tend to take more time to play, a lot of post-jam voters will not even play the game. An example would be "The Stanley Enigma"[66], a Ludum Dare 28 entry by Amos Wenger and Amadine Bru (Fig. 3.3). Though it is not a pure text-based game, it contains 1K+ lines of dialogue, 90% done by Amandine Bru, because Amos Wenger had to program the whole game[67]. The comments were not bad, but as described in the Pre-Production chapter 5.1, the comments are often superficial.

3.1.4 Story Exploration Game

With the recent success of Dear Esther [7], Gone Home [8], or Firewatch [9], a new genre has arisen, called Story-Exploration games. The goal of those games are generally to explore a world and to discover a story through this exploration.
The difficulty with this genre of game for a game jam is to define its scope and to shorten the story to its essential parts. Taking a game that seems

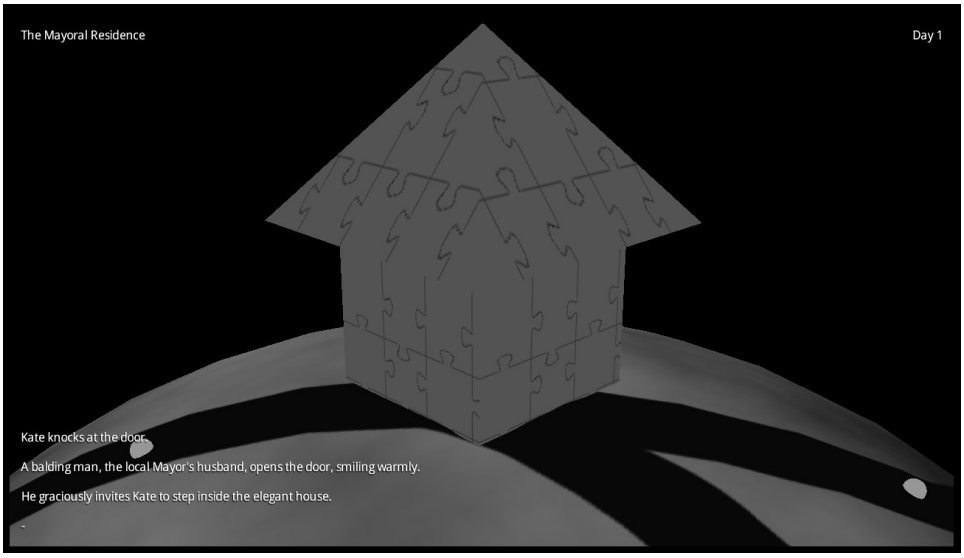


Figure 3.3: A night in a house in the game "The Stanley Enigma", a Ludum Dare 28 entry by Amos Wenger and Amadine Bru.

short like "Dear Esther", the remake of the mod started around May 2009 [68] and finished the February 14th 2012 [69]. A game of this scope cannot be done in a weekend. Like for point and click adventure, the development team has to focus on creating small levels and a short story. For example, the game jam entry "The Last Promise" (Fig. 3.4) has four different levels that the player walks through with four lines of dialogue for each of them.

3.2 Narrative in a Mechanic game

Most of times, game jam entries tend to be mechanic-oriented games. This tend to create games that have a very good short-term motivation, but fail on the middle-term motivation pretty quickly.
In a game jam, the work intensity varies a lot depending on the role of the jammer. An artist will generally work a lot at the beginning of the game jam and the intensity will go down, due to the fact that the programmer needs time to implement anything that the artist is doing. On the other hand, the programmer will see his work intensity grow during the time span of the game jam. This symmetry can be used to add narrative at the end of a game jam. The characteristic of the narrative tools have to be easy to be implemented by the programmer and have an important narrative impact on the game.
Wilhelm Österberg defines eleven narrative tools in his Master Thesis about "Storytelling in Single Player Action Computer Games"[58], but only several can be used in the span of the end of a game jam:

Figure 3.4: The main character walking in the sewer in the game "The Last Promise" created for the Ludum Dare 27 by the author.

- Pre-Rendered Video: It is pretty easy to implement for a programmer on a game engine that features video textures. For example, "Meow: the Great Starvation" (Fig. 3.5) implements pre-rendered video for the Game Over screen.

Figure 3.5: Meow, a hungry cat in the house of a stranger, facing a dog in the video-game "Meow the Great Starvation"

- In-Game Cutscenes: Unfortunately, compare to Pre-Rendered Video,

it adds a lot of time to implement In-Game Cutscenes for the programmer. They have to be planned at the beginning of the game jam.

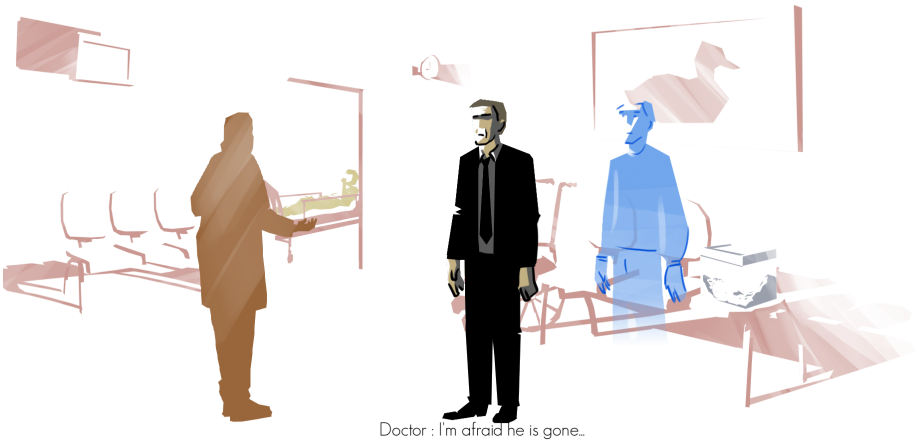


Figure 3.6: A blue ghost son and his father with a doctor in the Ludum Dare 29 entry "Final Move"

- Character Development: Making more than one stage is hard during a game jam, so character development, that requires several stages is a hard narrative tool to be used in game jam. The author's game "Final Move" (Fig. 3.6) has two stages that include the development of the main character.
- Voice-over: For a programmer, voice-over are mostly sound files, that have to be placed at the right time. Thus, they are easy to implement in the context of a game jam. For example, three games from the author use voice-over:
  - "Rise of Akeicha", a point-and-click adventure, uses voice-over for the main character in conjunction with subtitles.
  - "Final Move", a point-and-click adventure, uses voice-over for the main character, the doctor and the father (Fig. 3.6).
  - "I love you no matter what", a single stage point-and-click adventure, uses voice-over for the main character, the mother and the father (Fig. 3.1).
- Ambient Actions are actions performed by NPC (Non-Player Characters) between each other or reacting to an action from the Player

Figure 3.7: Akeicha on the desert that Earth have become from the "Rise of Akeicha", a Bacon Game Jam entry

Character. Creating NPC is already an enormous task for a game jam, making them react to each other is even more complex. In the author's game "Power Creep" (Fig. 3.2), villager turn horrified when the main character kill someone in front of them.

- Location Evolvement: Like "Character Development", the span of a game jam does not allow for a lot of diverse content. Using the same level and make it evolve explode the scope of content needed, even a simple shader make already harder to evolve a location.
- Interactive Dialogue: With a writer, a programmer can work together to create Interactive Dialogues and have them react to events. For example, a tool to make it work is UnityTwine [86] a Unity plugin that allow to use narrative multilinear trees from Twine.
- Storytelling Items allow to structure the narration with the gameplay goal or to not force the narration on the player. For example, in the author's game "I love you no matter what" (Fig. 3.1), the player clicks on several objects and can choose to discover the story choosing any items they want.
- Player Journal is not a useful narrative tool for game jam, because game jams entry are too short to make it work as intended in bigger production.
- Plain Text: Text or dialog are some of the simplest narrative tools that can be used in game jams. For example, the author's game "Black

Figure 3.8: The main character climbing a hill with his own thoughts in the game "The Last Promise" a Ludum Dare 27 entry by the author.

Whiskers" (Fig. 3.9) uses dialog to create the narration between the main character and the three bosses. An other example is "The Last Promise" (Fig. 3.8), where the thoughts of the main character are written as plain text subtitles.

3.3 Black Whiskers' Narrative Design Process

"Black Whiskers" (Fig. 3.9 3.4.8) was a Game Boy Jam 4 [87] (2.3.1) entry created by the author, where the player controls a pirate cat called Black Whiskers who avenges the treason of his ex-crewmen.
The only restrictions of the Game Boy Jam are the game resolution (160x144), the color palette of maximum 4 at the same time and input controls like the Game Boy. It occurred during 10 days, from August 8th - 18th 2015. Because there is no theme restriction, the brainstorming can happen before the game jam itself, making it a more gameful game jam, because participants can form team and prepare the game they want to create before the event takes place.
The game idea was to create a top-view hack'n'slash. To make the story as flexible as possible, Black Whiskers was fighting his ex-crewmen who betrayed him to get his revenge. The story could work with one to several bosses, depending on the available time. The basic structure for the dialogues were, a subtle title for the level presenting the boss to be encountered, the boss encounter dialogue, and finally the defeat of the boss dialogue.

Figure 3.9: Black Whiskers facing his last foe, the masked Big Boss, in the viedo-game "Black Whiskers" created for Game Boy Jam 4

As a Mechanic-based game, Black Whiskers needed time not only for the narration, but also for the gameplay implementation, increasing the scope of the project. In bigger project, there is already a struggle between narration and gameplay and what happens when you scale down the project is that the struggle becomes more intense. In a game jam context, only one of those two aspects can be perfectly implemented.
The Minimum Viable Product was a one-level game with a boss at the end having a dialogue intro. The initial boss was Krokoss, a crocodile with a cannon arm. As the minimum goal was mostly done on the 13th of August (Fig. 3.10), after the basics implementation of the game, the scope increased to three levels, a world map and three bosses. With two last days of crunch, the game could be released on the 17th.
The comments were mostly positive, except for the difficulty [5]. With so much time put in the creation of three different bosses and three different dialogues, with a narrative twist at the end, the gameplay suffered a lack of iterations that would have optimized the player experience.

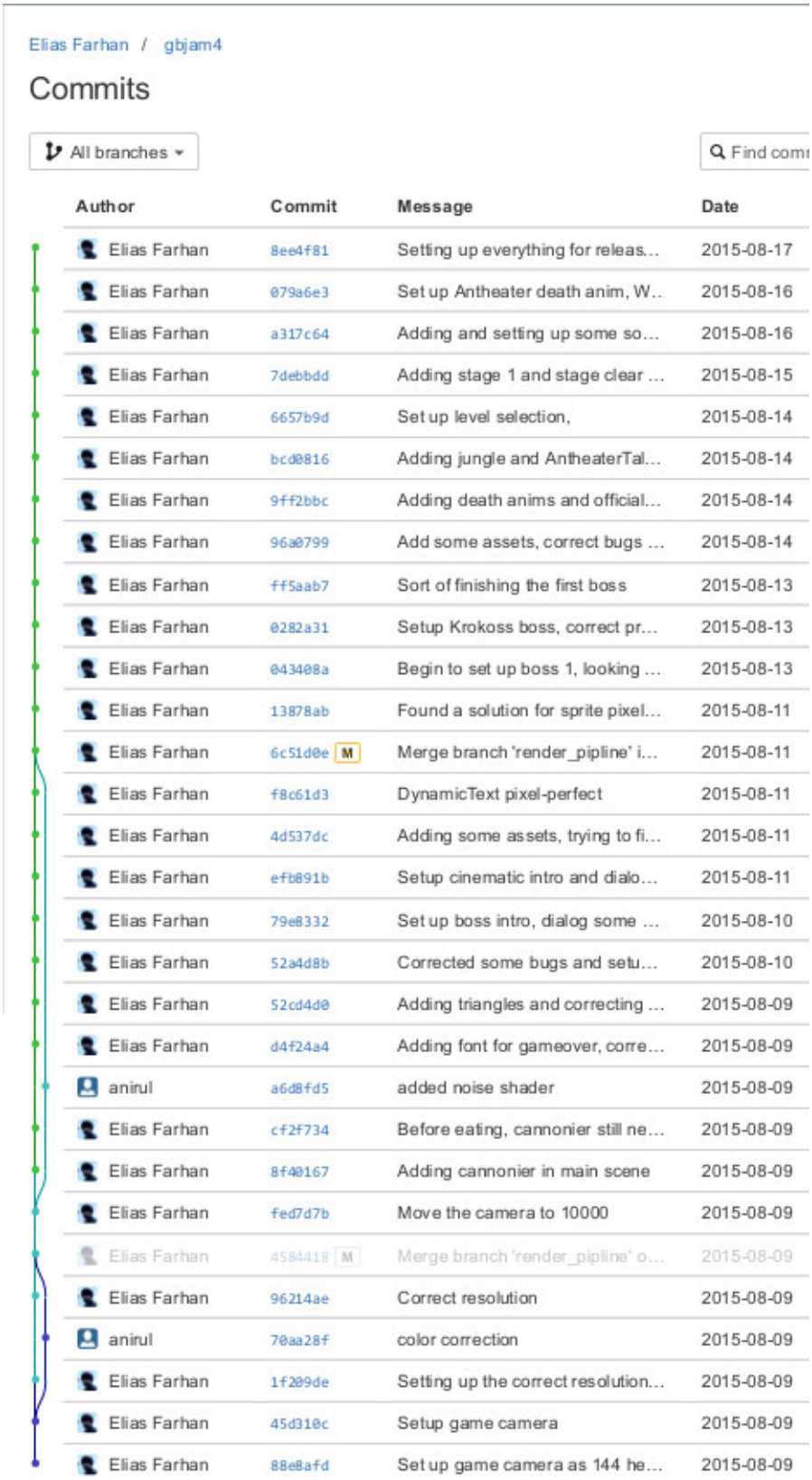


Figure 3.10: Second page of the author's git commits for "Black Whiskers", the Game Boy Jam 4 entry.

3.4 Author's Narrative Game Jam Entries

As part of the One Game A Month challenge [94], the author created in the past two years several narrative games during game jams. They are ordered chronologically.

3.4.1 The Last Promise

"The Last Promise" [19] is a story-exploration game jam entry done during the Ludum Dare 28, a 72-hour game jam, December 13th-16th 2013, with the theme "You only got one". The challenge was to create a narrative from scratch in term of codebase. The narrative tool used was plain text monologue.

3.4.2 Meow: the Great Starvation

"Meow: the Great Starvation" [21] is a infiltration game jam entry done during the Bacon Game Jam 7, a 48-hour game jam, March 21 - 23 2013, with the theme "hungry". One of the main challenges was to implement a cinematic in a custom engine as narrative tool.

3.4.3 Final Move

"The Final Move" [20] is a point-and-click game jam entry done during the Ludum Dare 29, a 72-hour game jam, April 25th-28th 2013, with the theme "Beneath the surface". The challenge was to implement a dialogue system from scratch and manage a team of eleven people, with voice-over actors as a narrative tool.

3.4.4 Furbrawl

"Furbrawl" [4] is a platformer created during the Game Boy Jam 3 [87], August 1st - 10th 2014, with no particular theme. The main challenge was to work continuously for one week long and implement a dialogue system in low resolution. The narrative tool used was plain text dialogue.

3.4.5 I love you no matter what

"I love you no matter what" [8] is a single room Point-and-Click adventure (3.1.1) created during the Ludum Dare 31, December 5th-8th 2014, with the theme "Entire Game on One Screen". The challenge was to use dynamic music transition working depending on the situation, with the custom dialogue system implemented already for "Final Move" (3.4.3). The narrative tool used was plain text dialogue with storytelling items played with voice-over.

Figure 3.11: The platformer "Furbrawl", a Game Boy Jam #3 entry

3.4.6 Power Creep

"Power Creep" [22] is a hack'n'slash sandbox created during the Global Game Jam, January 23 - 25 2015, with the theme "What do we do now?". The challenge was to create a big world in the scope of a game jam. The narrative tool of Ambient Actions, where villagers react to the murder of their friend, was a joke between the developers.

3.4.7 Rise of Akeicha

"Rise of Akeicha" [11] is a point-and-click adventure created during the Bacon Game Jam 9, June 13 - 15 2015, with the theme "Depth". The challenge was to reimplement the dialogue system into Unity and make it work with the voice-over. The narrative tool used was plain text dialogue with voice-over.

3.4.8 Black Whiskers

"Black Whiskers" [5] is a hack'n'slash created during the Game Boy Jam 4, August 8 - 18 2015, with no particular theme. The challenge was to create three different bosses and have a custom cinematic dialogue system

as narrative tool.

3.5 Conclusion

Game jams are solid events for beginner game developers who want to learn the art of game development. It allows them to understand the concept of scope.
Game jams often output arcade game, with very simple mechanics. Narrative design is a critical aspect to create compelling narration in game jam entry, even in more mechanic game.
In this game jam context, it is important to define the different narrative tools that can be implemented in any narrative game. One of the easiest solutions is to use plain text dialogue, with eventually voice-over, as they are easy to setup. Some narrative tools are useless like Player Journal due to the impact depending on the scope of the game or time-wasting like In-Game Cutscenes, that need a lot of implementation on the programmer side.
One of the main challenges for narrative games in game jam is the scope. The story needs to be flexible and it should be easy to cut a part of it due to the time-limit format. The definition of the Minimum Viable Product during the brainstorming phase (2.2.5) of the game jam is of high importance for the success of the implementation of the story.
Even with all the tools available in this chapter, the true problem of creating a narrative game during a game jam is the lack of depth possible to implement in those short time-format. As Chris Hecker rants [52]

> Game jams are shallow by design.

Game jams are probably not the best format to create compelling storytelling games, but they can give a good start as Pre-Production process (Chapter 5). Typically, if a one-room point-and-click adventure like "I love you no matter what" (3.4.5) is not fun or can not captivate the interest the player, a point-and-click adventure with a lot of rooms will not be more captivating, nor fun to play.

Chapter 4

Big Team Game Jam

4.1 Introduction

The event format that are game jams has grown in shape and size during the last decade. With record-breaking events like the Global Game Jam[16], the Ludum Dare[17] or the Nordic Game Jam[24], game jams are now mainstream game development events. Even the industry appropriates the format, like the Amnesia Fortnight [33], where Double Fine Productions employees worked on something different than their main project for two weeks. But in all of those game jams, the team size is always around one to ten people, sometimes a bit more with free agents[1] moving between several teams.
Expanding the team size means dealing with management and communication challenges. But as game jams tend to imitate studio practices visible in contemporary game development [2], game jams did not yet try to imitate the feeling of being in a big production team.
Switzerland, like a lot of different countries around the world, does not have a AAA-industry. For Swiss game developers and game designers to live those kinds of experience, they have to go work outside of Switzerland. In the context of the Zurich University of Arts (ZHdK)[30], students learn game design in a generalist approach, without any big production experience. On the other side of Switzerland, in the Swiss-French indie game developer's community, a lot of different game studios often work together in several local game jams every year.
In this Swiss context, a question arises: "How to bring all those different talents together to make them work on one game?". Both of those communities tried to answer this question with two different formats that are part of the overall definition that the authors call Big Team Game Jams. The Swiss-French community with the initiative of David Javet created the I^3 Game Jam, using the time-limit format and expanding the team size in a standard game jam. The Zurich University of the Arts with René Bauer ini-

tiated the Swiss Mercenaries Workshop where participants worked for three weeks on Swiss Mercenaries, a pre-defined game concept.
This paper will look at both those approaches and the authors will define the different parts that form Big Team Game Jams: Target Group, Hierarchy, Participants Motivation, Ownership, and Duration. The goal for this paper is for communities or schools in countries with no AAA-industry to have a framework to create such type of events at very low-cost.

4.2 Big Team Game Jam

In this section, the authors formalize the Big Team Game Jam framework and look at the different challenges that would arise compared to standard game jams. The goal of this framework is to give the opportunity to present the format to game jam organizers who want to create their own Big Team Game Jam.

Definition 9. *A Big Team Game Jam is a game jam with a team of more than 10 people, creating together exactly one game.*

Big Team Game Jams have a specific goal: to gather a lot of participants around one game project. With game jam as a mimesis of the game development process[2], increasing the team size also imitates the AAA industry practices and approaches.
In this context, the authors define game jams by their time-limit format between a couple of hours to one month[1], thus enabling other types of events like workshops that are not per se a playful game jam.

4.2.1 Target Group

Definition 10. *There are two categories of Big Team Game Jams:*

- *Indie Big Team Game Jams (I^3 Game Jam [9], Section 4.3.1)*
- *Academic Big Team Game Jams (Swiss Mercenaries [10], Section 4.3.2)*

A Big Team Game Jam always answers questions from a community with different motivations and needs. With the categories of Definition 10, the authors also define the goal of each of those events:

[1] A longer duration would define the event as a standard game development process.

Definition 11. *Depending on the target group, Big Team Game Jams have different goals:*

- *Indie Big Team Game Jams gather different game studios and developers together.*
- *Academic Big Team Game Jams bring big production experience to students.*

From Definition 11, because of Indie Big Team Game Jams' goal to gather people together, the setting of the event is local. People that would participate online would not have the same experience. Due to the local feature, it is hard to to gather otherwise employed people for the duration of a whole week, so the duration is the same as a standard game jam: 48 hours. It also means that Indie Big Team Game Jams are a challenging and regional affiliation type of game jams [3].
On the other hand, Academic Big Team Game Jams can take place in between semesters. The duration can be extended compared to Indie Big Team Game Jam. Due to the flexibility of duration, the challenge of the event decrease, making Academic Game Jam career advancement type of game jams, because students come for the experience to work on a bigger team.

4.2.2 Hierarchy

As standard game jams have smaller teams, organization in a game jam team is not a main challenge of those events. One quality that is found in game jams is that the participants keep their individuality in the group and have ownership of the final product[1].
Compared to standard game jams, Big Team Game Jams ease up the team formation process. Because people are separated in the different specializations, the forming process is Supervised Group Forming [4]. It becomes vital to understand the skills of everyone in the team and, compared to standard game jams, people with similar skills will work together instead of trying to go to different teams.
With Big Team Game Jam, middle management or a lead team is a necessity. The notion of equality and individuality is lost to the team spirit. The lead team has the decision power over the game production. Even in this industry-like context, it is important to emphasize the motivational drive:

Definition 12. *Like in standard game jams, participants in Big Team Game Jams have a voluntary and intrinsically motivated commitment [1].*

From Definition 12, it is also important to define that participants can leave the game jam when they want.
Even if individuality is lost, the notion of ownership is important to be defined. With so many people involved in the same project, giving complete ownership to every participant is a blocking situation in case of conflict. There are several solutions:

Definition 13. *The ownership of the game product of a Big Team Game Jam has three different categories:*

1. ***Jam Organization Ownership**. The game product is owned by the game jam organization and any attempt to sell the game will give the revenue to the game jam organization.*
2. ***Participant Ownership With Jury Condition**. The game product is owned by the participants, but any attempt to go outside of the jam context has to be authorized by the game jam organization.*
3. ***Full Participant Ownership**. The game product is owned by everyone and everyone has the right to block any attempt to go outside of the jam context.*

4.2.3 Participants' Motivation

According to the authors' observations, like in standard game jams, participants come with different motivations and expectations.

Definition 14. *The motivations of the participants to Big Team Game Jams are categorized like this:*

1. *Active participation in the community*
2. *Attachment to the game idea*
3. *Learning big production experience*

4. Knowledge exchange

These motivations are not dependent on the categories of the Big Team Game Jam (Definition 10). The active participation in the community motivation arises when the Big Team Game Jam targets a specific community. The notion of individuality in a Big Team Game Jam comes from the attachment to the proposed game idea. If people can criticize the game idea, then attachment can be a greater motivation, compared to a dominant lead team forcing the game idea. Knowledge exchange and big production experience are two different things. Knowledge exchange is between the participants as the big production experience is between the participants and the organization.

4.2.4 Team Selection

Definition 15. *The teams of Big Team Game Jams have two different selection processes:*

- *Invitation Team Selection*
- *Open Team Selection*

The selection processes from Definition 15 are independent from the categories of Definition 10.
Invitation Team Selection processes are more closed and controlled selection processes, providing a more gameful experience[1]. It allows to limit the number of participants, and creates a sensation of being part of an elite. It gives more motivation for participants, but also creates a separation in the target developer community between those who were invited and the others, making it, according to René Bauer, a personal affair between the producer and the participants. The team forming is dependent on the selection process, which thus affects the hierarchical structure (see also 4.2.2). Invitation Team Selection is more prone to mistakes than Open Team Selection, because the producer and the lead team can invite people they think are friendly, but are not competent or not able to integrate the team. Therefore, a producer has the choice between quality and comfort, if he for example choses a team member who is competent but finds it hard to integrate himself in the group.
Open Team Selection processes are more open, providing a more playful experience[1]. The filtering process can still occur during the game jam

when participants that can not integrate themselves into the team have to leave. It balances the unfairness from the Invitation Team Selection by making the selection process open to everyone, but it also removes a certain value to the event from the "I am invited" attitude.

4.2.5 Duration

Definition 16. *Big Team Game Jams last from 2 days to 1 month. Their duration can also be extend to several iterations.*

Duration becomes important to discuss during the organization of a Big Team Game Jam. In an Indie Big Team Game Jam (Definition 10), participants work in their day job, making it hard to expand the duration for more than 48 hours. On the other hand, due to the team size, the organization of the team occuring at the beginning of the event takes a longer time, making it harder to make a game jam of several hours. For example, a game jam like the 0hr Game Jam[36] would be impossible in a Big Team Game Jam context.
In Academic Big Team Game Jam, the duration can be extended if students are on holidays, making it possible to last during several weeks.

4.3 Practical Cases

The authors' definition of Big Team Game Jams from Section 4.2 comes from two practical examples that occured in Switzerland during the year 2015. Based on interviews with the two organizers, the authors analyzed the two structures of the two events to formalize the Big Team Game Jams.

4.3.1 The I^3 Game Jam

- First edition: 28-30 July 2015
- Second iteration: 28 November 2015
- Third iteration: 16-17 January 2016
- Fourth iteration: 27 February 2016

Figure 4.1: The Swiss-French team of the first I^3 game jam. Courtesy of the I^3 Game Jam [9]

Swiss-French Context

The Swiss-French game developer's scene revolves around a monthly meet-up. People from Lausanne, Geneva, Fribourg, Neuchâtel, and Wallis meet there and participate together in various local game jams (for example in 2015):

- The Swiss Castle Jam, that is the Swiss-French Global Game Jam (2.3.2) location [11].
- The LevelUp Game Jam (2.3.2), located in the Engineering School of Fribourg [12].
- The Epic Game Jam (2.3.2), occurring during the NIFFF (Neuchatel International Fantastic Film Festival) [13].
- The Kid Game Jam (2.3.2), where kids were creating the game concept and the assets [14].
- The Level Design Game Jam (2.3.2), sponsored by Nintendo being the first game jam on Super Mario Maker [15].

In this playful context, the game developers' community is not supported by any engineering or design school in term of specialization, making the meet-up and game jam vital for the community.
Inspired by this atmosphere, the core idea of the I^3 game jam was to gather 26 Swiss-French game developers (Figure 4.1) and create a game AAA-style in 48 hours. More than this, the concept of the game jam was for

the participants to exchange their knowledge and to test the mix of game development philosophies between all the different game studios on a large scale. One implicit goal at the time was to create a local multi-player game for the SwitzerLAN contest for Swiss game designer. The ownership of the game product was to everyone. If someone wanted to continue to work on the game product outside of the I^3 Game Jam context, the totality of the revenue would go to the I^3 Game Jam, making it a Jam Organization Ownership (Definition 13).
Compared to standard open game jams, to control the chaos that would arise from bigger teams, the participants were selected and invited by David Javet, the project manager, and the number was limited to 25, making it a Invitation Team Selection (Definition 15). This made the event a more personal initiative, thus private. One of the fear of David Javet, who is also the organizer of the monthly meet-up, was to destroy the community due to the clashes that could occur during the event, due to the number of different personalities or working philosophies. But it was something that David Javet wanted as an social experience, and the motivations for the participants were to be active in the community and learn from each other (Definition 14).

Pre-Production

The brainstorming and initial game concept was done by a core lead team composed by David Javet (Producer), Aurélien Da Campo (Lead Programmer), Mathieu Bonvin (Lead Artist), and Sandro Dall'Aglio (Lead Designer) on the Thursday evening. The theme that was given was "42". The number was separated to have a 4-player local multi-player game, where 2 players were required to win.
For David Javet, the link as winning condition could also be used as a way to advertise the concept of the jam as a bridge between the different Swiss-French game studios. The game idea was then presented to the whole team on Friday evening with the lead team having to defend their ideas. For David Javet, the lead team came too lightly prepared on Friday evening and they really had to fight for the game idea against the individual expectations of the whole team.

Production

The production went on from Friday evening to Sunday evening. The team was divided in two: the programmers and the artists. The programmers had a strict sleeping schedule, with, for example, Aurélien Da Campo forcing the programmers to sleep the Friday night at 1:00AM (very similar to advices from Kaitila's Game Jam Survival Guide[5]). On the other hand, the artist worked during the first night, falling asleep at 7:00AM, making the

programmers wait on Saturday for them to wake up at 2:00PM. In terms of team space distribution, the programming team and the artistic team were clearly separated in two different rooms.
One of the main challenges was communication. Rumors began to arise. David Javet gives two examples during his interview. During a coffee break, he heard artists saying that there was no more wall-jump in the game. Of course, he directly reacted by reaffirming that wall-jump was still a feature of the game. An other anecdote was that a programmer named a frontal attack a 'dash' and as the rumor began to spread, artists finally were drawing a dash attack instead of a normal frontal attack. The job of producer was a constant verification of the team workflow and team understanding.

Outcome

At the end of the event, the game was playable, but the choice was taken to make four more iterations, with the promise for the community to show the game at festivals like the Stunfest [32] and the Ludicious [31]. The next iterations had the same format, with a more open and voluntary approach. Participants from the first edition were not forced to come back at the next iterations, making it possible to invite new members of the community.

4.3.2 The Swiss Mercenaries Workshop

- First edition: 1-21th August 2015
- Second iteration: 1-12 February 2016
- Third iteration [in planning]: 1-21 August 2016

Swiss-German Context

The Zurich University of the Arts (ZHdK) [30] has a game design specialization since 2005. A lot of successful Swiss game projects come out of the school. Graduate students go working on the small local game industry or create their own indie studio, with very rare exceptions going to the AAA-industry. René Bauer, the Head of the Master program, describes in his interview the formation as a generalist one, with game development in small teams, similar to standard game jams.
The inspiration for the workshop theme came from the American game industry taking inspiration from the European mythologies and history (for example the God of War serie is inspired by Greek mythology). The Swiss

Figure 4.2: The Swiss Mercenaries Workshop announcement logo. Courtesy of the Swiss Mercenaries [10]

Mercenaries universe is national Swiss history from the Middle Age. The requirement was that it could be can developed over several series and genres. The lead team chose hack'n'slash as the game genre, but taking inspiration from Assassin's Creed, René Bauer still thought it was not mainstream enough. The goal of the workshop was not to be personally linked to the game idea, but for the students to live the big production experience (Definition 14).

Pre-production

The lead team began to work on a prototype three weeks before the workshop, such that the workshop was a production and not a pre-production process. In term of gameplay, René Bauer was searching for a simple hack'n'slash mechanic, not as complex as modern fighting mechanics like Shadow of Mordor. The goal of this pre-production process was to have a perfect prototype, but the core mechanics did not fulfill his expectations.
From the beginning, the vision was to make several iterations of the workshop, so there was not as much pressure like in the I^3 Game Jam to finish something at the end. The innovation of the game was to make an animation-based hack'n'slash.

Production

In terms of structures, the team was divided in six departments: visual, programming, sound, narration, level design, and game mechanics. The duration of the overall workshop production was three weeks, meaning that people could work from home. In term of proportion, the visual team was mostly working from home, compared to the six other departments. For

this to work, the team used Trello & Slack as tools to raise a team spirit awareness and for communication and management.
René Bauer saw that there were communication problems between the level design and the programming team, that were expressed because of the space distribution of the workplace. The programming team was to the extreme right of the room, the narrative & game mechanics in the middle and the level design team to the extreme left. For René Bauer, the level designers & programmers should sit next to each other.
In term of knowledge exchange, compared to the I^3 Game Jam, participants were not supposed to develop any new skills during game development, but to work on what they already know, with a focus on specialization, removing the knowledge exchange motivation from Definition 14. On Trello, people had to take job that they were able to do.
On the overall experience, René Bauer says that a lot of students and people loved to work on their specialization (as ZHdK [30] have a generalized game design curriculum). He was astonished by how many assets the team could create, but he was still disappointed that the game mechanic was not working at the end of the workshop like he expected to. Slack was a great tool for the team spirit. As a critic, he thinks that those sorts of events need a "punishing producer" who pushes the participants and enforces the original Big Team Game Jam idea, otherwise the participants will work on what they want.

Outcome

Two new workshops on the same game subject were decided to complete the game and to arrive at a point were the leadership of the project would decide if they wanted to go on with the project, similar to the I^3 Game Jam post-event process.
In term of ownership, any participant can take the project and produce it himself, but with the authorization of the lead team after showing a development plan, for example to raise a Kickstarter campaign, making it a Participant Ownership with Jury Condition (Definition 13).

4.4 Discussion

During the Ludicious Zurich Game Festival [31], René Bauer and David Javet were invited to present and to discuss the two approaches each community took. During the discussion, they were comparing the two models.
Using Table 1, the two models can easily be compared in the context of the Big Team Game Jam framework (Section 4.2). An interesting row is the Promotion. David Javet promotes more the event as a link for the Swiss-French game development community rather than the game that they created. This is comparable to promotion around standard and mainstream

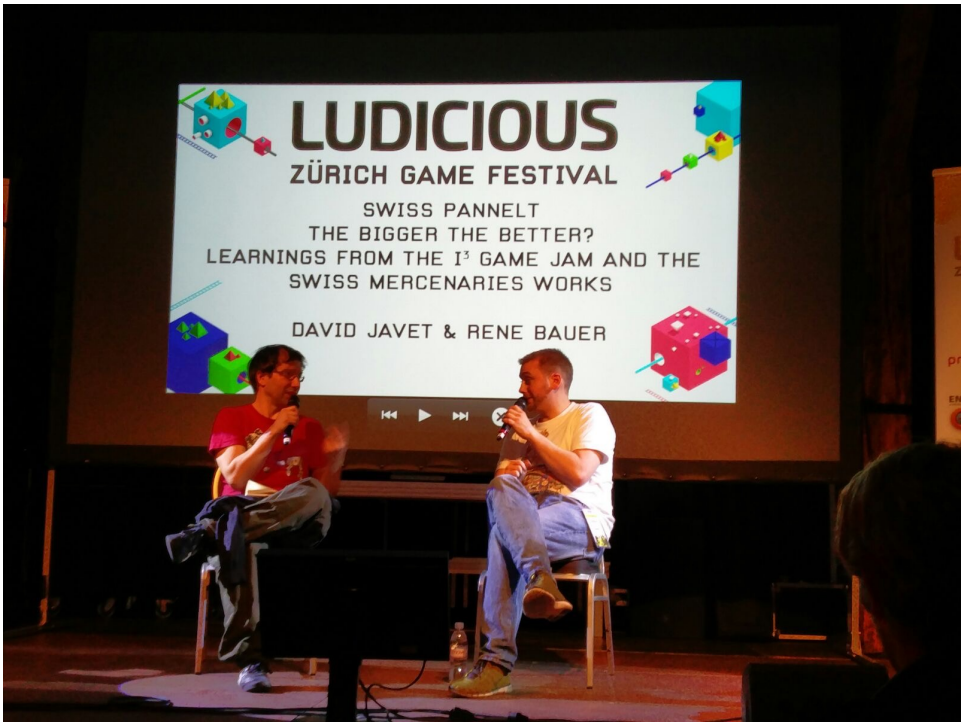


Figure 4.3: René Bauer (left) and David Javet (right) discussing about Big Team Game Jams during the Ludicious Zurich Game Festival.

game jams, where the event is largely exposed compared to the games created in the event, with some few exceptions. René Bauer focuses more on the game concept and final game, as the workshop name already contains the name of the game concept.
The Duration row is a clear practical reflection of the Definition 16 and the challenges between Indie Big Team Game Jams and Academic Big Team Game Jam.
The Participants' Motivation row was the hardest to define (Definition 14). The I^3 Game Jam was clearly linked to the Swiss-French game developers' community. Meaning that working on the I^3 Game Jam was practically translated to active participation in the community. Due to the format near standard game jams, the I^3 Game Jam participants still had their words to say on the game idea, making a feature of the *attachment to the game idea*. Like David Javet said in his interview, the lead team had to fight for the game concept. Finally, David Javet also enthuses the educational part of the I^3 Game Jam, because any participant could learn from the methods and working philosophies of the game studios participating in the event.
On the other hand, the Swiss Mercenaries has not a pre-defined game idea and is not linked to a community. René Bauer in his interview clearly in-

Table 4.1: Categorization of both events

	I^3 Game Jam	Swiss Mercenaries
Event type (Def. 10)	Indie BTGJ	Academic BTGJ
Part. Motiv. (Def. 14)	1, 2, 4	3
Ownership (Def. 13)	1	2
Promotion	The Event	The Game
Duration (Def. 16)	48 hours	3 weeks

sisted that the participant were using their skills and were working on tasks they were able to complete - thus making the main participants' motivation the experience of working on big production.

4.5 Conclusions

During the Ludicious Q&A [31], the producers of two game jams, David Javet and René Bauer gave several advices for organizers of such events. David Javet advised not to rush in the event and to create a solid communication structure, and René Bauer advised to not be afraid to fail.
The framework that the authors offer allows countries with no AAA-industry or universities who want to give the big production experience to their students, to create Big Team Game Jams with a theoretical basis to enable the organization discussion and to evaluate the different options given to them. Like democratized mainstream game jams, Big Team Game Jams appeal to a range of different motivations and skills for participants at very low cost. The current format that gives this framework also gives the possibilities to expand the game product outside of the jam with a clear understanding of the notion of ownership.
The Big Team Game Jams framework clearly defines the different challenges for events that want to gather a lot of developers together. Two models arise depending of the target group of the event: Indie Big Team Game Jams and Academic Big Team Game Jams (Definition 10). Those two categories have their specifications and own goals.
Knowledge exchange and active participations (Definition 14) between game studios are the main features of Indie Big Team Game Jams. On the other hand, the big production experience is the main feature of Academic Big Team Game Jams, as they target students. Other kinds of motivations like active participation in the community or attachment to the game idea are elements that can be used to emulate individuality and more enthusiasm.
Ownership (Definition 13) is a very important subject not to underestimate and to make all the participants aware of, because conflicts can occur around the property of the final game.
Team are to be selected carefully as participant should be competent and

be able to integrate the whole team (Definition 15). David Javet even goes forward on prioritizing Comfort (meaning integration) over Quality (meaning competence). More open team selection allows a better filter process in case of conflict.
Finally, the two Swiss Big Team Game Jams were a success at their initial goal and, with the work of the authors, allows now to export a new game jam format at very low-cost for countries without any AAA-industry.

Chapter 5

Game Jam as Pre-Production Process

According to the "Findings from the Global Game Jam 2014 six-month follow-up survey" [19], 35% of participants continue to work on their games after the end of the game jam. A lot of different successful games were game jam entries. As more and more game jams are becoming popular among the game industry, more and more commercial games start as game jam entries. With this new way of starting a game project, it is important to define what are the differences with already known pre-production processes, like for example prototyping. For this, we bring contents from two conferences into this chapter:

- GDC Next 10, 2013, Nathan Vella: *From Game Jam to Full Game* [46]
- GDC Europe 2014, Mattia Traverso and Alan Zucconi: *From a Cookie to the Full Cake - Redesigning 'game jam' Games* [47]

As a pre-production process, game jams' entries are different in kind and scale, compared to commercial and released products. Game jams' entries are a lot more dependent to their game jam's context. By definition, game jams are fixed in time, but a game project is not. Transitioning from a fixed timespan to a more flexible one is a well known challenge.
For this chapter, the author will take the development of the video-game Project Splash (originally called "Super Splash Fisticuffs" [10] and now called "Splash Blast Panic" [35]) as an example. It was originally a Ludum Dare 32 entry with the theme: "Unconventional Weapons".
Project Splash is a local multi-player competitive party game, where players fight each other using a water gun and flying with a jet-pack. The core concept were discussed in the train to Basel in 15 minutes. The idea was to create a party game with a cartoony feel, which also means non-violent losing condition. The "fight" is about pushing the opponents out of the

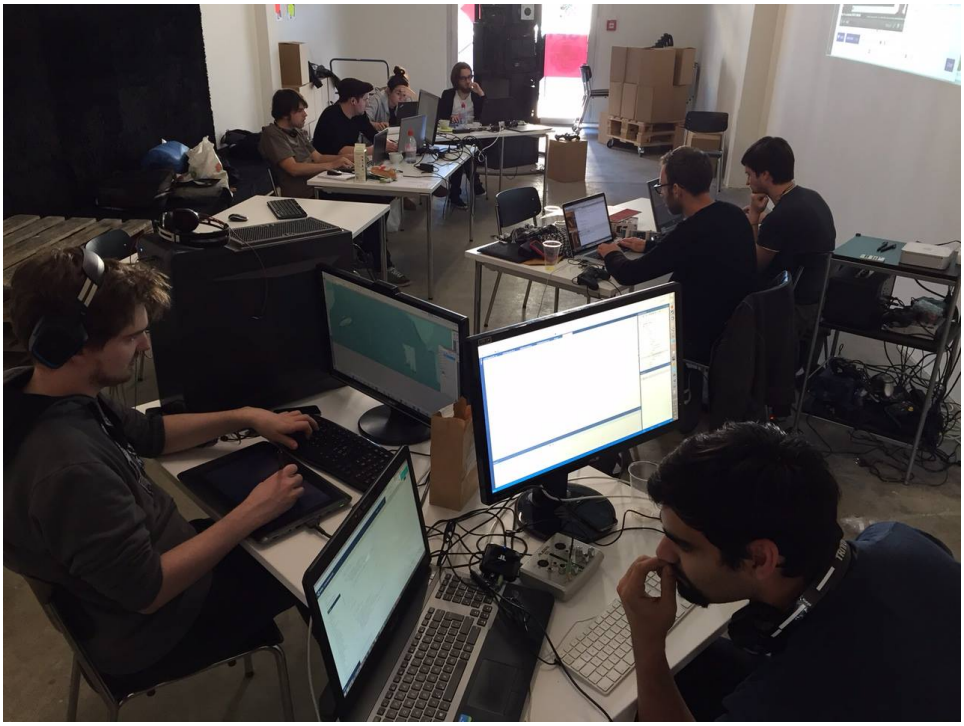


Figure 5.1: A programmer & an artist jamming during Ludum Dare 32 @ Spielhalle Oslo, Basel.

stage, not killing or hurting.

Figure 5.2: Super Splash Fisticuffs, Project Splash basis, after the Ludum Dare Jam 32.

The Project started at Ludum Dare 32 and continued throughout the second half of 2015 & the first half of 2016.

5.0.1 Game Jam differences with Prototyping

Taking the process of game development that can take some years and squeeze it to order of magnitude of days have several consequences on how the product will look like. From Nathan Vella's conference [46], a game jam entry is about "finishing something" and a prototype is about "answering a question", but the two of them is "about an idea, then getting a controller in someone's hand to get them playing that idea".
For the author, game jam is more about creating a complete product, as prototyping is about testing an idea.
The timespan of a prototype is also not limited. In a prototyping session, the product defines the prototyping design method, compared to the time-limit format that defines a game jam (Definition 1). In a typical game development process, a playable version can come after two weeks. This order of magnitude is already too big for a game jam. Playable version comes directly and as soon as possible.

5.0.2 Fun Time

A standard prototype being in the context of a longer timespan, a game jam has to be redesigned. What Nathan Vella [46] names the "Fun Time" goes from minutes (max. 15 min.) for a game jam entry to 1-2 hours of fun. In 2013, when Nathan Vella gave his talk [46], it was rare to have a place for a released game under one hour, but now it is even harder with the 2-hours refundable limit from Steam. Of course, Steam is not the only distribution platform, but even consoles are not a good market for less than one hour life expectancy for a game.
This requirement of longer Fun Time influences the setup of the production (discussed in 5.2). As a commercial product, it has to compete with other games in the genre, which is not the case of the game jam entry who compete with entries of the same game jams.
Of course, this requirement for a successful commercial product is not mandatory for a non-commercial end-product.

5.0.3 Context of a Game Jam entry

During GDC Europe 2014, Mattia Traverso and Alan Zucconi made a talk together about the process of redesigning a game jam entry. Both were in the production of a used-to-be game jam entry.
Paraphrasing Mattia Traverso, a game jam has a context. For example, when it is a local game jam, people are joking about the theme, creating

inside joke not understandable for the external world. Included in a fully released product, those jokes can not work as intended.
From this point, Mattia Traverso says that some game jam entries are not translatable to a fully released game. A good example of this is the Alt Ctrl Game Jam (2.3.1), where participants have to experiment with unusual input controls. If a participant creates a custom controller with Arduino and duct tape, he will not be able to distribute it, except if he commercializes the controls, which is unlikely.
Of course, most of the game jams' entries are translatable to full release products. The main question is then "how?". This question relates also to the context of the game jam. What is still going to be in the final game and what is going to be cut?
As pre-production process for Project Splash, the core concept was linked with the theme "Unconventional Weapons", but could easily go out of the jam itself. Even with the redesign, the core concept stayed in the game, and people playing now the demo do not know where it comes from.

5.1 Post-Jam Measure of Success

Definition 17. *Online ratings and comments are shallow compared to physical play-testing session.*

Most of the famous game jams have a way to vote online for the entries. For example, Ludum Dare (2.3.1) has a three weeks voting period after the event week-end. The point is that the exhibition of the game jam entry is often part of the game jam.
Evoland [6] won the Ludum Dare 24 Compo category [92]. After this initial success, Nicolas Canasse decided to setup the production of a full commercial experience. The main problem with the online voting is that it is superficial compared to a physical play-testing session. Also, the game designer does not control the environment in which the tester plays. Going to voting period of Project Splash, the entry had 15 comments:

- 8 comments praised the art.
- 4 comments were about controls restriction problem.
- 4 comments said the game was polished.
- 1 negative comment on the overall fun.

The Ludum Dare 32 had 2821 entries. One can not expect voters to play more than five minutes an entry. Some players take the time to record themselves playing and/or stream themselves, but in the end, it becomes really important to make a physical play-testing session. A great example of this would be a convention or a festival.

Figure 5.3: The setup of Project Splash during Fantasy Basel

Super Splash Fisticuffs was shown during Fantasy Basel 2015 (Figure 5.3). Controllers were at disposal and players could just take them and play. Watching the players' reactions was already a better feedback than any comment from the game jam page.
The second time the author really showed the game was the first milestone tournament. The goal was to see if the redesign of the gameplay was working through a play-testing session disguised as a tournament with prizes. The tournament was a success and we could gather a lot of data and feedback, which would have come much more later in a standard development process.

5.2 Setting up production

If game jams are used as a special pre-production process, it is important to understand what are the challenges and solutions to go from a pre-production to a production process.

Definition 18. *The author defines three types of post-jam production process:*

- *Content expansion*
- *Features addition*
- *Playful iterations*

5.2.1 Content Expansion

Definition 19. *The Content Expansion Post-Jam Production is a process where the developers expand the already implemented contents.*

Alan Zucconi [47], taking his game Orbitalis as an example, explained the principle of "Expand the content = Expand the Experience". He takes the game Titan Souls [3] as an example for the same methodology. The basic gameplay of Titan Souls did not change since the Ludum Dare [93], where it was voted the best game of the jam. Instead, they just added other bosses to expand the experience. He adds that this approach is not perfect by also stating that it is not sure that a successful game jam entry can make a successful released game. On the other hand, he also states that ambition costs a lot because: "The more time you develop your game, the more copy you have to sell". So, in the context of a successful game jam entry, a good

methodology is to expand the content. Nathan Vella [46] also asks what should be expanding: contents, features, or scope?

5.2.2 Features Addition

Definition 20. *The Features Addition Post-Jam Production is a process where the developers add new features that were not thought in the initial game jam concept idea.*

Mattia Traverso [47] on the other hand compared to Alan Zucconi goes the otherway around: "I don't jam to sell my games." His approach can be interpreted as prototyping an idea and then redesigning around the answer that the prototype asks. The example he takes is the game Thomas Was Alone [11], where there is a huge difference between the game jam entry and the final version. The addition that Mike Bithell made is a voice over narration, which became the core idea of the game, compared to the original version where it was just text and the focus was on making rectangles jump. He defines his methodology as: "Not being trap in the original vision."

5.2.3 Playful Iterations

Definition 21. *The Playful Iterations Post-Jam Production is a process where the production schedule follow the same pattern than a game jam.*

During the Q&A of his presentation at Ludicious 2016 [31], Nicolas Canasse defines his post-jam production of Evoland [6] like "jamming".
The setup of production for Project Splash was a chaotic process. Originally, the game jam product was not intended to be continued. We had some good feed-back during the game jam by other physical participants that had fun with the game, but the voting was not successful. We brought the game to Fantasy Basel without second thought and players had a lot of fun, on the game jam product. The decision was then made to set up full-production and the first milestone was a private tournament with prize. Of course, scheduling was not our strength and the full production began one week before the tournament. We managed to have a new fully functional version for the tournament with some minor bugs that were not in our goals of the tournament.
During the redesign, the core design stayed the same, but instead of a flux of water, water gun were now shooting at a certain frequency. Other strategies were added, like the dash down, items, and bouncing. So, overall, we

stayed true to the vision, but changed the mechanics to increase "Fun Time" through skill learning.
The progress of the project can be seen after one year of production (Fig. 5.4). The developers schedule every two-week Friday to work on the game, adjusting the mechanics, updating the graphic assets, and stabilizing the gameplay. The author had the possibility to go at the GDC and to show the game to game developer around the world and even to take contact with Sony of America to port the game on their platform. One of the main chal-

Figure 5.4: Players fighting in the Winter level in Splash Blast Panic, one year after the start of the production.

lenge that Nathan Vella [46] did face during the production of Super TIME Force was motivation. For a game jam, it is easy to find the motivation for a 2-days production, but it is another problem for a long-term development. For Nathan Vella, the key is good scheduling.

5.3 Industry Game Jam

Before going into this section, it is important to define several categories of game jams

Definition 22. *"Elite" Game Jams are closed game jams, with selected participants.*

Definition 23. *Commercial Game Jams are "Elite" Game Jams with a focus on making the Game Jam an entertainment product.*

This section will focus on industry game jam [1]. Those game jams target professional for them to try something different from their normal workflows. Nathan Vella quickly give some advice for those "Commerical" Game Jams at the end of his presentation [46]:

- Longer duration
- Clear goal
- Far less scale issues

5.3.1 Amnesia Fortnight

Figure 5.5: Tim Schaefer speaking to his employees during the Amnesia Fortnight. Courtesy of Double Fine [97].

This topic is the occasion to talk about Double Fine Amnesia Fortnight [33]. It is a game jam where all workers from Double Fine Productions [97] could propose an idea, then people on Internet voted for their best ones and finally, teams formed inside Double Fine Productions developed in two weeks, four to five of the best ideas developed into games. A lot of successful products came out of those Amnesia Fortnights and those events brought creativity in the studio during crisis times.

The first Amnesia Fortnight (2007-2008) was a private event that did not raise exposure and can be categorized as an "Elite" Game Jam (Definition 22). It occurred during the tense development of Brütal Legend [12]. Tim Schafer decided to split his 60-person development team into four groups, challenging them to create short but complete games in just two weeks [34]. But the two next editions (2012 [60] & 2014 [62]) were open to the public on the Internet to vote on the idea and to follow the process.
Compared to the Global Game Jam (2.3.2), the Amnesia Fortnight does not ask its participants to release the code source of their game, because they are property of Double Fine. With the invitation-only participation and the recording feature, the Amnesia Fortnight is a Commercial Game Jam (Definition 23).

5.3.2 MOJAM

Before the Amnesia Fortnight 2012, Mojang did the MOJAM [61], a 60 hours game jam in cooperation with Humble Bundle for charity. They livestreamed the whole development process, making it a Commercial Game Jam, like Amnesia Fortnight (Definition 23).
In 2013, they invited other indie game studios to join them for MOJAM2 [63], including Oxeye Game Studio, Wolfire Games, Vlambeer, Grapefrukt Games, and Ludosity. For example, Vlambeer created Wasteland Kings (Fig. 5.6), an action rogue-like like, that became Nuclear Throne, a successful commercial game.

Figure 5.6: Wasteland Kings, a MOJAM 2 entry by Vlambeer. Courtesy of Vlambeer [64].

5.3.3 "Game Jam" Reality Show

The most expensive game jam (costing about $400,000 to produce according to) was a complete disaster and stopped after one day of shooting [91] in

Figure 5.7: The setup of the "Game Jam" reality show, with some Mountain Dew branding in the background. Courtesy of Robin Arnott and Polygon [90]

April 2014. The goal was to show "showcasing the magical world of indie game development to a public audience."
The problem was that "[...]the show devolved into a reality-style contest". For Robin Arnott [90]: "We all understand it was first a show and second a game jam.", but when a producer asked him: "Do you think you're at an advantage because you have a pretty lady on your team?", it was too much. The consequence was that all the developers left the show and killed the production. Like the author described (6.2.1), game jams are a collaborative process, not a mere competition. It is easy to understand that the production of the show thought that they could adapt a competitive format to this new activity that was game development and game jam. The result was an clear answer, that would be confirmed in the context of the #GamerGate controversy, that was occurring later in 2014, but started already in February with the Steam Greenlight process of Depression Quest, a game by Zoe Quinn, also a participant to the "Game Jam" reality show.

Chapter 6

Game Jam as Design Method

Game Jams have a very great future in front of them. They are events that boost a community into active cooperation. The Swiss-French game development community (4.3.1) is a perfect example of that, with five official local game jams each year, and many more unofficial ones.
Game jams are a Game Design method without any doubts. Widely spread and known around the community, game jams have also been used in the game industry by company like Double Fine Amnesia Fortnight(5.3.1). To be defined as a design method, it is important, not only to define the process of a game jam (like in 2.2), but how the process is used.
This chapter tries to merge the results from the different chapters in a more design point of view, leaving the world of game design. The author already defined game jam as event with product-based goal (Definition 1), but on this chapter, he describes his design process for the Exhibition Game Jam and the two approaches to game jam, first as collaborative, and then as a performance.

6.1 Author's Participation to Game Jams

With an autoethnographic approach, the author participated in a lot of different types of game jams that are listed in Table 6.1.
An interesting aspect of this table is that the games' titles, the game jams' names and the themes were mostly known by heart. Except some extraordinary events that occurred during game jams, a participant mostly remembers the product and the structure of the events. Gorm Lai [98] on why he did not continue to work on a game jam entry after the deadline explained that game jams are more about the process. Though the game jam process (shown in Table 2.3) is clearly defined, it is extraordinary moments that come to mind. For example, from the Epic Game Jam #2 (2.3.2), the author remembers the theme revelation through the destruction of a pinata (Fig. 6.1).

Figure 6.1: The pinata's destruction as theme revelation for the Epic Game Jam #2. Courtesy of the Epic Game Jam [13].

The author tried to participate in one game jam each month, as part of the One Game A Month challenge [94]. An interesting aspect of this process was for the author to have a different team for each of those game jams, getting to know people from all around the world.

Table 6.1: Chronological Author's Participation to Game Jams Table

Game Jam Name	**Game Title**	**Date**	**Theme**	**Team Size**
Bacon Game Jam 6	Trials	25-27.10.2013	Rainbow	3
Ludum Dare 27	The Last Promise	13-16.12.2013	You only got one	5
Cyberpunk Jam	STRIDERS	12-24.03.2014	Heights	5
Bacon Game Jam 7	Meow: the Great Starvation	21-23.03.2014	Hungry	4
Ludum Dare 28	Final Move	25-28.04.2014	Beneath the surface	11
Epic Game Jam 1	Sea Mark	04-06.07.2014	First time	4
Game Boy Jam 3	Furbrawl	01-10.08.2014	*None*	5
Bacon Game Jam 8	MAD	17-19.10.2014	Millions of them	4
Post-Human Jam	Thanks for your cooperation	28-30.11.2014	Post-human	4
Ludum Dare 31	I love you no matter what	05-08.12.2014	Entire game in one screen	6
Global Game Jam 2015	Power Creep	23-25.01.2015	What do we do now?	5
Nordic Game Jam	Wash Machine Simulator	06-08.02.2015	Obvious	3
Ludum Dare 32	Super Splash Fisticuffs	17-20.04.2015	Unconventional Weapons	3
Bacon Game Jam 10	Rise of Akeicha	12-14.06.2015	Depth	4
Epic Game Jam 2	Meow Zedong	05-07.07.2015	90% water 10% evil 5% skin	5
Game Boy Jam 4	Black Whiskers	08-18.08.2015	*None*	5
Kid Game Jam	Fire Monsters	24-25.10.2015	*None*	3+5 kids
PROCJAM	Catpcha	07-16.11.2015	*None*	2
Ludum Dare 33	Fat Cat Rock and Roll	11-14.12.2015	Two Buttons & Rolling	1
Global Game Jam 2016	The Witch, The Irish, and The Imp	29-31.01.2016	Ritual	4
0th Swiss Train Jam	Meow Railways	14.02.2016	Rail	2
Indie Train Jam	Runaway Train, L4z0r 54mur41, and Escape the Party	10-12.03.2016	Maximum Capacity	5

6.2 Exhibition Game Jam

Coming from the very playful Swiss-French game development community (4.3.1), the author's exhibition game jam in Zurich had to disrupt the format of game jam [45]. In the design process of the game jam, thoughts were about creating a performance of game jam, with the author experience. From "Living the indie life: mapping creative teams in a 48 hour game jam and playing with data" [57] talking about game jams:

> This improvisational performance is all about those making the games: the people and the principles of their creative process. [...]
>
> Game jams, for us, are performances of design thinking, a style of creative thinking-in-action where ill- defined problems necessitate empathy and creativity.

And Schön says about improvisation:

> Improvisation consists on varying, combining and recombining a set of figures within the schema which bounds and gives coherence to the performance. ([59], p. 55)

Trying to make a solo improvisational performance for the author was not successful. A practical experiment was tried during the Ludum Dare 33. The results were a little prototype and without the group pressure, the author decided to cancel his participation, not satisfied with the result [75]. The main reason was the lack of group pressure.
From this experiment, the idea was to open the game jam to everyone, even non game designer or game developer. The idea switched from the two extremes that are solo and participative performance. In the same idea than happening [8], but still staying in the philosophy of a game jam, the exhibition game jam should help people create assets for a merged game.
A prototyping phase allowed the author to imagine the jam space (Fig. 6.2). The jam space has a bed and a big central table with several computers for people to use. The room has three walls:

- The Design Wall is about game ideas.
- The Motivational Wall is about motivation phrase, compliments, kind words for and from the participants.
- The Critics Wall is about judging design ideas and game jam.

Though the final place for the Exhibition Game Jam did not allow to use real walls, the ideas is to reflect, during the game jam, the work of the game jam on the three walls.

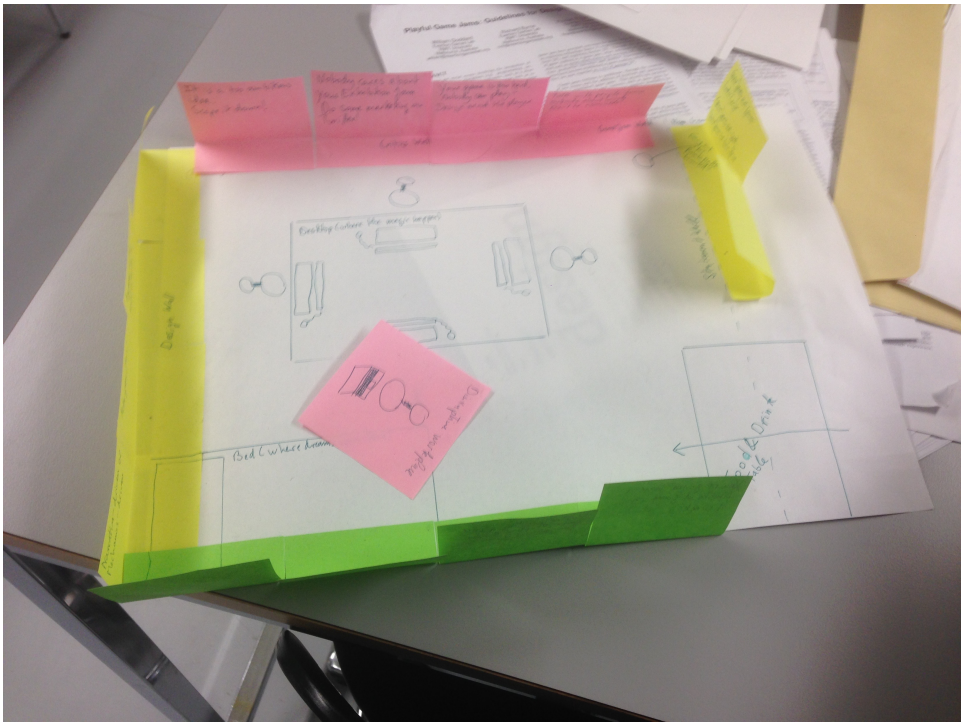
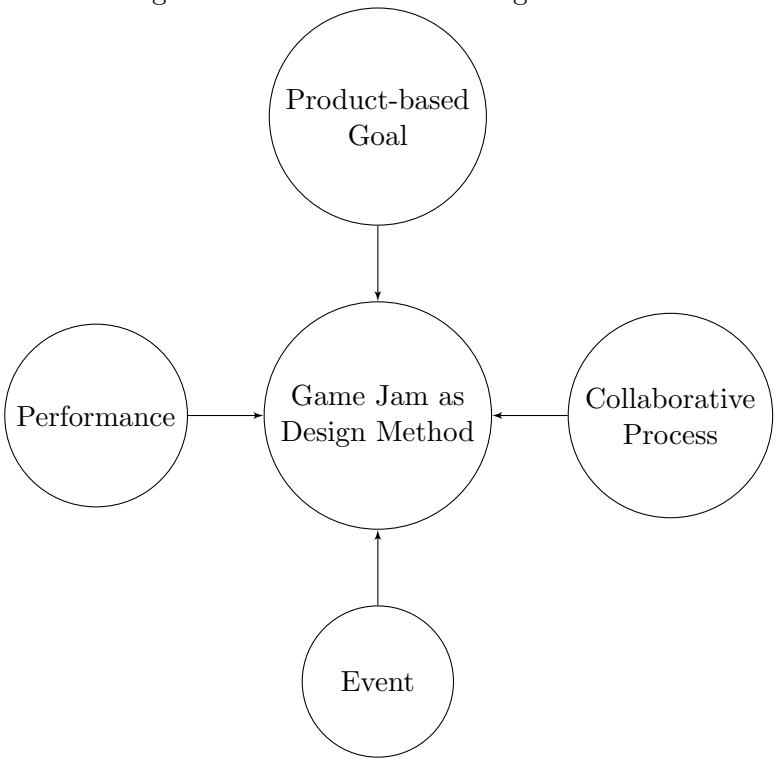


Figure 6.2: Prototyping of the Exhibition Game Jam, with the three walls of Motivation, Design and Critics

The final idea is the Frankenstein Game Jam, where participants create assets during two days and then, as a performance, the game developer merges all those assets in a single game. The result is a Frankenstein creature game, merged from several art styles, visions and ideas of the final product. Inspired by the author process during the Kid Game Jam (2.3.2), where children were creating the assets and then the game developer created the video-game from them. The difference is that, for the Frankenstein Game Jam, the author has the design decision control, asking people to create certain assets, though they keep their playful freedom to do so or not, compared to the Kid Game Jam where children had the design decision control, guided by the game developer on what was possible in a week-end.
This final project for the Exhibition Game Jam allows the same event to cover two different points of view on game jams. The first part where open participants come to create assets is a collaborative process. The second part where the game developer, alone, create the game from the assets is a performance.

Figure 6.3: Game Jam as Design Method



6.2.1 Game Jam as Collaborative Process

Jammers come together during a game jam to create a game. Like written in the paper "Playful Game Jams" [1]:

> Game jams are social events, often with over 50 participants at an event (e.g. fab48hr, LD48, GGJ, HBF). The coming together of many different people proves a strength for game jams by facilitating discussion and multi-disciplinary collaboration.

Game jams' team have a common goal to create and finish a game in the timeframe of the event. This collaboration works on a multi-disciplinary level and helps participants discover and learn skills from other discipline. The open participation values from game jams, that are less used for Elite Game Jams, helps a democratised game king process regardless of their professional role and at any level of education.
Gorm Lai [98] goes even further by praising the fact that the Global Game Jam (2.3.2) forces developers to release their code source for knowledge sharing.

6.2.2 Game Jam as Performance

Jammers perform a creative action during a game jams. This creative action is extraordinary, making it memorable. The first reaction to people outside of the game design world to game jams is astonishment about the timeframe. "How can you create a game in two days?" Of course, the knowledge of such people are that video-games last several hours, a task impossible to realize during a game jam where a fun time of fifteen minutes is already an accomplishment.
A well known format is the Ludum Dare Compo category (2.3.1). Though the experience of solo performance was not successful for the author for the Ludum Dare 33 Compo. After the ideation process, the execution of the Fat Cat Rock and Roll idea did not satisfied the author who cancel his participation to the game jam [75].
The participation of the author to the One Game A Month challenge [94] is a performance on its own. Even if the author participated in most of his game jams in team, game jams can be a team-based performance.

6.3 Critics of Game Jam

Game jams are accepted in the overall game developers community. Anchel Labena, one of the organizer of the Nordic Game Jam (2.3.2), even promotes game jam as a way to enter the industry [54].
As the game jam model is now defined, it is also important to look at his weaknesses. A good counter-argument is that game jam due their scheduled

nature and theme constrained can restrict artistic liberty. Folmer Kelly in his blog post on Gamasutra [50] sums the difference between doing a Game Boy themed game alone and during Game Boy Jam (2.3.1) by: "Someone else told me when to do it."

6.3.1 Depth Jam Design Process

The Depth Jam created by Jonathan Blow [51] and Chris Hecker [52] was a design response to several types of events, like game jams:

Figure 6.4: The 0th Indie Game Jam (March 15th to 18th 2002 2.3.2). Courtesy of the Indie Game Jam [38]

"Game jams are shallow by design." (Chris Hecker [52]). The Indie Game Jam (IGJ 2.3.2) was founded on the idea of exploring the design ramifications of a crazy technical question. For example, the first IGJ (Figure 6.4) restriction was to use 100,000 sprites in a custom game engine created by several participants[1] before the game jam. For Jonathan Blow, in contrast, contemporary game jams are more open. "It's great to make a game, any game." For him, this community spirit is often upheld as the best part of a game jam. For experienced developers, these jams are not so good, due to an overall social context that supports stagnation. "You are still part of

[1] Chris Hecker, Sean Barrett, Doug Church, Casey Muratori, and Jonathan Blow

a community that likes you because you are participating in a game jam." He says that the game developer communities must take care to separate the notion of participating in the community from the notion of success. He thinks that would be a nice start if there were a category of game jams with higher expectations. Though the IGJ could be included in this category, he does not think it is the right thing going into the future, because the time-limited format prevents the participants from going deep.
From this critic, Jonathan Blow and Chris Hecker proposed a new format, called Depth Jam, for on-going project, mixing ideas from several genres of events: game jams, conferences, meet-ups, and retreat-like events. The idea was to have 4 developers, talking about their games for 4-day and trying to give feedback to each others.

6.3.2 Why I'm no longer participating in game jams

Folmer Kelly [50] begins his blogpost by quoting moshboy announcement of a new game jam: "I have been vocally against game jams in recent months. They are held with such frequency and so commonly at the same places that they mostly taste vanilla and rarely feel special."
Folmer Kelly agrees with moshboy that there are indeed too many jams, that did not really factor into the Folmer Kelly's decision to stop taking part in them. Looking at his pro-jam friend, Folmer Kelly felt that there was a level of attachment to the concept of game jams in his arguments that Folmer Kelly found interesting. It felt like his pro-jam friend perceived: "I'm against jams", the same way as if someone had said: "I'm against more people making game."
Are we perpetruating the idea that game jams make games happen rather than people make games happen? Folmer Kelly started feeling like game jams have become a forced frame for creativity, a required activity. "You wanna make games? Do jams".
Folmer Kelly asks himself the question:

> "Would I have made less games if I hadn't participated in those jams? Knowing myself, the answer is "probably not". Would I have made better games if I hadn't participated in those jams? Possibly, since I wouldn't have put myself in a position where I was stuck with a theme and a deadline I didn't set myself. Did participating in jams get me more exposure? Participating in jams and using the associated hashtag will get you noticed by other participants which can lead to new fans, but on the flipside, odds are that the jam as a whole will get more exposure than any individual entry.
>
> So what is the difference for Folmer Kelly between making a GB style game and participating in GBJam? Someone else told me

> when to do it. Game jams are a valuable tool, but with the abundance and popularity of game jams, we're at risk of losing a sense of spontaneous creativity."

Former Kelly finishes his blog post with this phrase: "We don't need jams to create, we need people to create."

6.4 Future of Game Jams

The always growing number of participants of the Global Game Jam (Fig. 2.1) shows a bright future for game jams. For example, 2016 was the biggest Global Game Jam to date with over 36'000 participants across 93 countries that created almost 6800 games [72]. As a spontaneous movement and as a improvised performance [45], different kinds of game jams will continue to expand. For example, the Indie Train Jam (2.3.2) puts an indoor and static event like a game jam and make it move through the United States of America. The author's exhibition game jam, called Frankenstein Game Jam (6.2) is also an example of disrupting the formula of game jams, with new ways of jamming.
With the Depth Jam experience, it is important to see where are game jams heading to, but also to see what kind of other game design methods will arise from the game jam movement. Depth Jam being more a retreat-like event like the Stugan [73].
With the I^3 Game Jam, the Big Team model have been put in the test of different methodology. The first edition was a big more playful as the second edition will be oriented to a more agile method development, thus more gameful. Completed by the Swiss Mercenaries(4.3.2) workshop that uses the same format, without the standard 48-hour time-format, game jams can be used in country without AAA-industry to still create high-quality games in short timespan.
"Elite" Game Jams also commercialize the notion of game jams as a entertainment show. Amnesia Fortnight (5.3.1) has sold the idea of game jams to non game designers gamers. The "Game Jam" reality show (5.3.3) disaster showed the limit of entertainment using notion of competition in a collaborative process like game jams.
Game jams will continue to expand to new formats and to create new design movements, bringing playfulness and creativity in the game industry, and allowing game developers to come out of their daily routines.

Chapter 7

Conclusion

Game Jams are an event-based design method, with a focus on experimentation, disruption and creativity. Trying to define game jams is as hard as trying to define what a game is. From Definition 1, the author clearly begin with the time-limit format of a game jam as the basis for the definition.
History is a very important of every medium and the game industry always had problems archiving a lot of different useful datas. Game jams are not clearly represented on Wikipedia [83], with examples like the Global Game Jam Wikipedia page [85] or the Nordic Game Jam Wikipedia page [84] that do not list every editions of each events. The author proposes a chronology of game jams (2.1) with the most important dates.
As event, game jams follow a certain process. Before the event, organizers come together to decide on the different constraints and the flow of the events. Those different decisions can be categorized into organizing philosophies and they influence at the end the playfulness of the game jam.
On the other hand, the participants, knowing the several restrictions, come to the game jam with expectations and a certain intrinsic motivation or as the paper on Playful Game Jam [1] explains that, "play in game jams has been fostered by voluntary participation and intrinsic motivation". This notion of ludic craft defines game jam as a playful design method.
During the game jam, teams are created using supervised or unsupervised team formation processes. Teams that are finally created come together to brainstorm a game idea that is manageable in the duration of the game jam event. Finally, most of the time of the game jam event is spend producing the game jam entry.
As soon as the game is release, people can judge the game and can be ranked accordingly. Of course, not all game jams propose a post-jam process with voting and judging. After that, depending on the success of the game jam entry, the developers can decide to continue the development, and then use the game jam process as a pre-production process.
Game jams' entries are different than prototypes. They are complete prod-

ucts. Nathan Vella [46], and Mattia Traverso and Alan Zucconi [47] talked about this new challenge, to take a game jam entry and make it a full commercial product. The first thing that comes to mind is the challenge to increase the fun time (5.0.2).
The author defines three different production setups after a game jam as pre-production process (5.2): content expansion is just about adding more assets without changing a lot of the gameplay, features addition is about adding new mechanics and changing dramatically the game, and playful iterations is about developing the game like a game jam.
Game jams have already been used as a pre-production process in the industry. The most concrete example is the Amnesia Fortnight (5.3.1), with successfull project like Spacebase DF-9 [13], Hack'n'Slash [14], or Stacking [15] being game jam entries and then transformed as full commercial products. Those industry game jams are more closed than the indie game jams like Global Game Jam or Ludum Dare, and their duration are often longer. They also tend to focus on entertaining a public to the game jam process.
In this same genre of Industry game jams, came from Switzerland the format of Big Team Game Jam (4) in 2015. Two events tried to answer the question "How can we bring all the talents of the region and made them create a game?". In the Swiss-French community, the I^3 Game Jam brought 25 developers for 2 days to create a game. In the Swiss-German community, the Swiss Mercenaries workshop brought 26 developers divided in 5 departments to create a game in two weeks. The author analyzed those two events and defined the Big Team Game Jam that was presented during the International Conference on Game Jams, Hackathons, and Game Creation at the David Brower Center of UC Berkeley in California in March 13, 2016 [96].
The author participated in several types of game jams, trying to create games of different genres. One issue he had was to create compelling narrative games. After several attempts, he defined a framework to understand that kind of narrative game genres (3.1) are possible to create during a game jam and what kind of narrative tools (3.2) can be used efficiently .
Finally, to define game jams as design method, the author had to understand the principles around design methods. They help the design process in several phases, defined differently by several authors.
The author approach to define game jam as design method is to divide them in two categories: game jam as collaborative process (6.2.1) and game jam as performance (6.2.2). He designed the Frankenstein Game Jam as part of the Zurich University of the Arts Design exhibition (6.2).
In the end, game jams are a game design method, that would be beneficial to be used as a design method for other area, like industrial design or architecture, outside of the burden of commercial pressure.

Chapter 8

Credits

The first people that I want to thank are my parents, Regina Fischlin and Saaid Farhan. They supported me financially and they always motivated me to follow my dreams. I will never be able to pay this spiritual debt that I owe them.
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Bibliography

[1] Goddard, Byrne W., Mueller R., F. F. 2014. Playful Game Jams: Guidelines for Designed Outcomes. In *Proceedings of the 2014 Conference on Interactive Entertainment*, (New York, USA, December 02 - 03, 2014), ACM, Article 6, 1-10.

[2] Locke R., Parker L., Galloway D., Sloan R., The Game Jam Movement: Disruption, Performance and Artwork. In *Workshop Proceedings of the 10th International Conference on the Foundations of Digital Games* (Pacific Grove, California, Asilomar Conference Grounds), June 22 - June 25, 2015.

[3] Fowler A., Lai G., Khosmood F., Hill R. "Brad". Trends in Organizing Philosophies of Game Jams and Game Hackathons. In *Workshop Proceedings of the 10th International Conference on the Foundations of Digital Games* (Pacific Grove, California, Asilomar Conference Grounds), June 22 - June 25, 2015.

[4] Pirker J., Voll K., Group Forming Processes - Experiences and Best Practice from Different Game Jams. In *Workshop Proceedings of the 10th International Conference on the Foundations of Digital Games* (Pacific Grove, California, Asilomar Conference Grounds), June 22 - June 25, 2015.

[5] Kaitila C. *The Game Jam Survival Guide*. Packt Publishing Ltd, 2012.

[6] Kaitila C. *Infographic: Survey Results* [Accessed 16.05.2016]: `http://ludumdare.com/compo/2011/12/28/infographic-survey-results/`

[7] Kaitila C. *Infographic: Game Jam Postmortems* [Accessed 16.05.2016]: `http://mcfunkypants.com/2011/infographic-game-jam-postmortems/`

[8] Kaprow, A. *Happenings in the New York Scene. 15-26 in Essays on the Blurring of Art and Life*. University of California Press, 1961.

[9] I^3 Game Jam. 2015. [Accessed: 16.05.2016]: `https://twitter.com/i3gamejam`.

[10] Swiss Mercenaries Workshop. 2015. [Accessed: 01.03.2016]: http://swissmercenariesgame.com/.

[11] Swiss Castle Jam. 2016. [Accessed: 01.03.2016]:
https://globalgamejamsuisse.wordpress.com/.

[12] LvlUp Game Jam. 2016. [Accessed: 01.03.2016]: http://www.lvlupgamejam.ch/.

[13] Epic Game Jam. 2015. [Accessed: 01.03.2016]: http://epicgamejam.com/.

[14] Kid Game Jam. 2015. [Accessed: 01.03.2016]:
http://www.kidgamejam.com/.

[15] Level Design Game Jam. 2015. [Accessed: 01.03.2016]: http://www.leveldesigngamejam.com/.

[16] Global Game Jam. 2016. [Accessed: 01.03.2016]: http://globalgamejam.org/.

[17] Ludum Dare. 2015. [Accessed: 01.03.2016]: http://ludumdare.com/compo/.

[18] Alt Ctrl Game Jam website, [Accessed 16.05.2016]: http://altctrlgamejam.com/

[19] Global Game Jam Research Community, *Findings from the GGJ six-month follow-up survey*, October 2014, [Accessed 16.05.2016]: http://globalgamejam.org/research

[20] Archive of the Ludum Dare forum after the Ludum Dare Zero, May 2002 [Accessed 16.05.2016]:
https://web.archive.org/web/20020609233501/http://ludumdare.com/articles/?link=v&arid=29

[21] Results of the Ludum Dare 18, August 2010 [Accessed 16.05.2016]:
http://ludumdare.com/compo/2010/09/06/ludum-dare-18-results/

[22] Indie Train Jam webiste [Accessed: 16.05.2016]: http://trainjam.com/about

[23] Gameloading: Rise of the Indies movie website [Accessed: 16.05.2016]: http://www.gameloading.tv/

[24] Nordic Game Jam. 2015. http://nordicgamejam.org/. [last retrieved: 1. March 2016]

[25] Nordic Game Jam 13 winners [Accessed: 16.05.2016]: http://nordicgamejam.org/2013/01/winners-ngj13/

[26] Lithuanian Game Jam. 2002. [Accessed: 16.05.2016]: http://ltgamejam.org/

[27] 1st Lithuanian Game Jam page, [Accessed: 16.05.2016]: http://ltgamejam.org/2002/index.php

[28] 2nd Lithuanian Game Jam page, [Accessed: 16.05.2016]: http://ltgamejam.org/2003/index.php

[29] Lithuanian Game Jam 2009 page, [Accessed: 16.05.2016]: http://ltgamejam.org/2009/

[30] Zurich University of the Arts. 2015. [Accessed: 16.05.2016]: https://www.zhdk.ch/.

[31] Ludicious Zurich Game Festival. 2016. [Accessed: 16.05.2016]: http://ludicious.ch.

[32] Stunfest. 2015. [Accessed: 16.05.2016]: http://stunfest.fr.

[33] Amnesia Fortnight. 2014. [Accessed: 16.05.2016]: http://www.doublefine.com/fortnight.

[34] Tim Schafer interview, gamespot, [Accessed: 16.05.2016]:
http://www.gamespot.com/articles/how-amnesia-fortnight-saved-double-fine/1100-6365180/

[35] Splash Blast Panic website [Accessed 16.05.2016]: http://playsbp.ch

[36] 0hr Game Jam. 2015. [Accessed: 16.05.2016]: http://0hgame.eu/.

[37] The Indie Game Jams listing website, [Accessed 16.05.2016]: http://www.indiegamejams.com/

[38] 0th Indie Game Jam, [Accessed: 16.05.2016]: http://indiegamejam.com/igj0/index.html

[39] 1st Indie Game Jam, [Accessed: 16.05.2016]: http://indiegamejam.com/igj1/index.html

[40] 2nd Indie Game Jam, [Accessed: 16.05.2016]: http://indiegamejam.com/igj2/index.html

[41] Caillois R., *Des jeux et des hommes*, Gallimard, 2nd edition, 1967.

[42] S. Rogers, *Level Up! The Guide to Great Video Game Design*, John Wiley & Sons, 2nd edition, 2014.

[43] B. Hanington, B. Martin, *Universal Methods of Design: 100 Ways to Research Complex Problems, Develop Innovative Ideas, and Design Effective Solutions*, Rockport Publishers, 1st edition, 2012.

[44] Jones J. C., *Design Methods: Seeds of Human Futures*, Wiley-Interscience, 1st edition, 1970.

[45] Locke R., Parker L., Galloway D., Sloan R., *The Game Jam Movement: Disruption, Performance and Artwork*, Proceedings of the 10th International Conference on the Foundations of Digital Games (FDG 2015), June 22-25, 2015, Pacific Grove, CA, USA.

[46] Vella N., *From game jam to Full Game - Super TIME Force & Other Shining Examples*, GDC Next 10, 2013, [Accessed: 16.05.2016]: `http://www.gdcvault.com/play/1019694/From-Game-Jam-to-Full`

[47] Traverso M. and Zucconi A., *From a Cookie to the Full Cake - Redesigning 'game jam' Games*, GDC Europe, 2014, [Accessed: 16.05.2016]: `http://www.gdcvault.com/play/1020879/From-a-Cookie-to-the`

[48] Hecker C., *Finish Your Game*, GDC rant, 2010, [Accessed: 16.05.2016]: `https://youtu.be/f6ta63pbc6Q?t=204`

[49] Crawford J., *Game Developer Magazine: Game Career Guide*, 's Article, 2014, [Accessed: 16.05.2016]: `http://dc.ubm-us.com/i/360245-game-career-guide-august2014`

[50] Kelly F., *Why I'm no longer participating in game jams*, Gamasutra Article, 2014, [Accessed: 16.05.2016]: `http://www.gamasutra.com/blogs/FolmerKelly/20140728/222032/Why_Im_no_longer_participating_in_game_jams.php`

[51] Blow J., *Opinion: The Depth Jam* , Gamasutra Article, 2012, [Accessed: 16.05.2016]: `http://www.gamasutra.com/view/news/171314/Opinion_The_Depth_Jam.php`

[52] Hecker C., *The Depth Jam* , Blog Post, 2012, [Accessed: 16.05.2016]: `http://chrishecker.com/The_Depth_Jam`

[53] Adams E., *The Designer's Notebook: How Should We Judge a game jam?* , Gamasutra Article, 2012, [Accessed: 16.05.2016]: `http://www.gamasutra.com/view/feature/162947/the_designers_notebook_how_.php?page=1`

[54] Labena A. , *Jamming your way into the industry*, Gamasutra Article, 2013, [Accessed: 16.05.2016]: `http://www.gamasutra.com/blogs/AnchelLabena/20131202/206018/Jamming_your_way_into_the_industry.php`

[55] Adamkiewicz A., *No, You Can't Make Video Games*, Gamasutra article, 2012, [Accessed: 16.05.2016]:http://www.gamasutra.com/blogs/AleksanderAdamkiewicz/20120703/173489/No_You_Cant_Make_Video_Games.php.

[56] Kultima A., *An Autopsy of the Global Game Jam 2012 Theme Committee Discussion: Deciding on Ouroboros*, Proceedings of the 10th International Conference on the Foundations of Digital Games (FDG 2015), June 22-25, 2015, Pacific Grove, CA, USA.

[57] Truna aka j. turner, Lubi T., Cameron O., *Living the indie life: mapping creative teams in a 48 hour game jam and playing with data*, in Proceedings of The 9th Australasian Conference on Interactive Entertainment: Matters of Life and Death, 2013.

[58] Österberg W., *Storytelling in Single Player Action Computer Games*, Master Thesis in Human-Computer Interaction, Media Technology Programme, at the Royal Institute of Technology, Stockholm 2006

[59] Schön, D. A. *The reflective practitioner: how professionals think in action*. New York, Basic Books, 2008.

[60] Humble Bundle Amnesia Fortnight 2012 page [Accessed: 16.05.2016]: http://blog.humblebundle.com/post/36081511427/do-like-double-fine-and-drop-everything-its-time

[61] Humble Bundle MOJAM page [Accessed: 16.05.2016]:

http://blog.humblebundle.com/post/17668356358/humble-bundle-mojam-this-weekend

[62] Humble Bundle Amnesia Fortnight 2014 page [Accessed: 16.05.2016]: http://blog.humblebundle.com/post/75815492301/the-humble-weekly-sale-featuring-double-fine

[63] Humble Bundle MOJAM2 page [Accessed: 16.05.2016]:

http://blog.humblebundle.com/post/43574635219/introducing-humble-bundle-mojam-2

[64] Wasteland Kings page, Vlambeer [Accessed 16.05.2016]: https://vlambeer.itch.io/wasteland-kings

[65] Wenger A., *Ludum Dare #29 Post-Mortem*, [Accessed 09.05.2016]: https://amos.me/blog/2014/ludum-dare-29-postmortem/

[66] Wenger A., Bru A., *The Stanley Enigma*, Ludum Dare 28 April 25th-28th 2014, [Accessed 09.05.2016]: http://ludumdare.com/compo/ludum-dare-29/?action=preview&uid=4598

[67] Bra A., *Ludum Dare 29 – Post Mortem*, [Accessed 09.05.2016]: https://thetoasterchronicles.wordpress.com/2014/05/06/ludum-dare-29-post-mortem/

[68] Briscoe R., *Mods and the Motherland!*, [Accessed 09.05.2016]: http://www.littlelostpoly.co.uk/mods-and-the-motherland/

[69] Dear Esther Steam Page, [Accessed 09.05.2016]: http://store.steampowered.com/app/203810/

[70] Global Game Jam Sets Guinness World Record™ for Being The Largest Game Jam in the World [Access: 26.04.2016]: http://2013.globalgamejam.org/news/2012/03/01/global-game-jam-sets-guinness-world-record%E2%84%A2-being-largest-game-jam-world

[71] A theory of designing, John Chris John, [Accessed: 13.05.2016]: http://publicwriting.net/2.2/theory_of_designing.html

[72] So long #GGJ16, and thanks for all the fun! [Access: 26.04.2016]: http://globalgamejam.org/news/so-long-ggj16-and-thanks-all-fun

[73] Stugan [Access: 26.04.2016]: http://www.stugan.com/

[74] Extra Credits, *Making Your First Game: Basics - How To Start Your Game Development*, [Accessed 16.05.2016]: https://www.youtube.com/watch?v=z06QR-tz1_o

[75] Farhan E., *I lost the game jam*, blogpost [Accessed: 01.05.2016]: http://eliasfarhan.ch/?p=393

[76] Post-Human Game Jam, Swiss-French game developers meet-up page, [Accessed: 16.05.2016]: http://www.meetup.com/Game-developers-Suisse-Romande/events/213793212/

[77] News Games Hackathon page [Accessed: 16.05.2016]: http://zhdkngh.tumblr.com/

[78] Darwinia Available Now On Steam [Accessed: 26.04.2016]: http://store.steampowered.com/news/493/

[79] iPhone 3G on Sale Tomorrow http://www.apple.com/pr/library/2008/07/10iPhone-3G-on-Sale-Tomorrow.html

[80] Cyberpunk Jam, 2014, [Accessed 16.05.2016]: https://itch.io/jam/cyberpunk-jam

[81] Finalist of the 1st IGF [Accessed 16.05.2016]: http://www.igf.com/1999finalistswinners.html#finalists

[82] Website for the 2006 edition of Nordic Game Jam [Accessed: 27.04.2016]: http://nordicgamejam.org/06/

[83] Wikipedia [Accessed: 16.05.2016]: https://en.wikipedia.org/

[84] Wikipedia Nordic Game Jam page [Accessed: 16.05.2016]: https://en.wikipedia.org/wiki/Nordic_Game_Jam

[85] Wikipedia Global Game Jam page [Accessed: 16.05.2016]: https://en.wikipedia.org/wiki/Global_Game_Jam

[86] Unity Twine plugin github repository[Accessed: 01.05.2016]: https://github.com/daterre/UnityTwine

[87] Game Boy Jam #4 [Accessed 09.05.2016]: http://jams.gamejolt.io/gbjam4

[88] Bacon Game Jam website [Accessed 16.05.2016]: https://bacongamejam.org/

[89] PROCJAM website [Accessed 14.05.2016]: https://itch.io/jam/procjam

[90] Campbell C., *How 'Game Jam,' an indie game dev reality show, collapsed on its first day of filming*, Polygon, Mar 31, 2014, [Accessed 15.05.2016]: http://www.polygon.com/2014/3/31/5568362/game-jam-reality-show-maker-studios

[91] Arnott R., *"GAME_JAM" and the Power of Integrity*, Gamasutra, 31.03.2014, [Accessed 15.05.2016]: http://www.gamasutra.com/blogs/RobinArnott/20140331/214349/quotGAMEJAMquot_and_the_Power_of_Integrity.php

[92] Evoland, Ludum Dare 24 page, [Accessed: 16.05.2016]: http://ludumdare.com/compo/ludum-dare-24/?action=preview&uid=8497

[93] Titan Souls, Ludum Dare 27 page, [Accessed: 16.05.2016]: http://ludumdare.com/compo/ludum-dare-28/?action=preview&uid=7984

[94] One Game A Month [Accessed 09.05.2016]: http://www.onegameamonth.com/

[95] Swiss Train Jam hashtag on Twitter [Accessed 16.05.2016]: https://twitter.com/hashtag/swisstrainjam

[96] The International Conference on Game Jams, Hackathons, and Game Creation Events website [Accessed 16.05.2016]: http://www.gameconf.org/

[97] Double Fine website [Accessed 16.05.2016]: http://www.doublefine.com/

[98] Lai G. Interview. 08.05.2016.

[99] Bauer R. Interview. 08.01.2016.

[100] Javet D. Interview. 09.01.2016.

Ludography

[1] The Walking Dead, 2012, PC, Telltales Games, Telltales Games [Accessed 10.05.2016]: https://www.telltalegames.com/walkingdead/

[2] Life is Strange, 2015, PC, Dontnod Entertainment, Square Enix [Accessed 16.05.2016]: http://lifeisstrange.com/

[3] Titan Souls, 2015, PC, Acid Nerve, Devolver Digital [Accessed 16.05.2016]: http://www.devolverdigital.com/games/view/titan-souls

[4] 0RBITALIS, 2015, PC, Alan Zucconi, Mastertronic, [Accessed 16.05.2016]: http://store.steampowered.com/app/278440/

[5] Super TIME Force, 2014, PC, Capybara Games, Capybara Games [Accessed 16.05.2016]: http://supertimeforce.com/

[6] Evoland, 2013, PC, Shiro Games, Shiro Games [Accessed 16.05.2016]:http://evoland.shirogames.com/

[7] Dear Esther, 2012, PC, The Chinese Room, The Chinese Room [Accessed 16.05.2016]: http://dear-esther.com/

[8] Gone Home, 2013, PC, The Fullbright Company, The Fullbright Company [Accessed 16.05.2016]: http://www.gonehomegame.com/

[9] Firewatch, 2016, PC, Campo Santo, Campo Santo [Accessed 16.05.2016]: http://www.firewatchgame.com/

[10] Nuclear Throne, 2015, PC, Vlambeer, Vlambeer [Accessed 16.05.2016]: http://nuclearthrone.com/

[11] Thomas Was Alone, 2012, PC, Bithell Games, Bithell Games [Accessed 16.05.2016]: http://www.mikebithellgames.com/thomaswasalone/

[12] Brütal Legend, 2013, PC, Double Fine Productions, Double Fine Productions [Accessed 16.05.2016]: http://www.doublefine.com/brutallegend/

[13] Spacebase DF-9, 2014, PC, Double Fine Productions, Double Fine Productions [Accessed 16.05.2016]: http://www.spacebasedf9.com/

[14] Hack'n'Slash, 2014, PC, Double Fine Productions, Double Fine Productions [Accessed 16.05.2016]: http://www.hacknslashthegame.com/

[15] Stacking, 2012, PC, Double Fine Productions, Double Fine Productions [Accessed: 16.05.2016]: http://www.stackingvideogame.com/

Author's Games

[1] Trials, Bacon Game Jam 6 [Accessed 16.05.2016]: https://bacongamejam.org/jams/bacongamejam-06/107/

[2] STRIDERS, Cyberpunk Jam [Accessed 16.05.2016]: https://itch.io/jam/cyberpunk-jam/rate/3946

[3] Sea Mark, Epic Game Jam #1 [Accessed 16.05.2016]: http://epicgamejam.com/game/sea-mark

[4] Furbrawl, Game Boy Jam 3, [Accessed 16.05.2016]: http://jams.gamejolt.io/gbjam3/games/furbrawl/31893

[5] Black Whiskers, Game Boy Jam 4, [Accessed 09.05.2016]: http://jams.gamejolt.io/gbjam4/games/black-whiskers/86604

[6] Millions of Aliens of DOOM, Bacon Game Jam 8, [Accessed 16.05.2016]: https://bacongamejam.org/jams/bacongamejam-08/287/

[7] Thanks for your cooperation, Post-Human Jam, [Accessed 16.05.2016]: http://gamejolt.com/games/thank-you-for-your-cooperation/40798

[8] I love you no matter what, Ludum Dare 31, [Accessed 16.05.2016]: http://ludumdare.com/compo/ludum-dare-31/?action=preview&uid=32076

[9] Wash Machine Simulator, Nordic Game Jam 2015, [Accessed 16.05.2016]: https://itch.io/jam/ngj15/rate/18561

[10] Super Splash Fisticuffs, Ludum Dare 32, [Accessed 16.05.2016]: http://ludumdare.com/compo/ludum-dare-32/?action=preview&uid=32076

[11] Rise of Akeicha, Bacon Game Jam 10, [Accessed 16.05.2016]: https://bacongamejam.org/jams/bacongamejam-09/421/

[12] Meow Zedong, The Great Helmsman, Future Chairman, The Urinal Leap Forward, Epic Game Jam #2,

[Accessed 16.05.2016]: http://epicgamejam.com/game/meow-zedong-great-helmsman-future-chairman-urinal-leap-forward

[13] Fire Monsters, Kid Game Jam, [Accessed 16.05.2016]: http://eliasfarhan.ch/?p=301

[14] Catpcha, PROCJAM 2015, [Accessed 16.05.2016]: https://itch.io/jam/procjam/rate/43630

[15] The Witch, The Irish, and The Imp, Global Game Jam 2016, [Accessed 16.05.2016]: http://globalgamejam.org/2016/games/witch-irish-and-imp

[16] Runaway Train, Train Jam 2016, [Accessed 16.05.2016]: https://itch.io/jam/train-jam-2016/rate/57573

[17] Escape the Party, Train Jam 2016, [Accessed 16.05.2016]: https://itch.io/jam/train-jam-2016/rate/57616

[18] L4z0r 54mur41, Train Jam 2016, [Accessed 16.05.2016]: https://itch.io/jam/train-jam-2016/rate/57647

[19] The Last Promise Ludum Dare page [Accessed 09.05.2016]: http://ludumdare.com/compo/ludum-dare-28/?action=preview&uid=32076

[20] Final Move, Ludum Dare 29 [Accessed 09.05.2016]: http://ludumdare.com/compo/ludum-dare-29/?action=preview&uid=32076

[21] Meow: the Great Starvation, Bacon Game Jam 7 [Accessed 09.05.2016]: https://bacongamejam.org/jams/bacongamejam-07/238/

[22] Power Creep, Global Game Jam 2015 [Accessed 15.05.2016]: http://globalgamejam.org/2015/games/power-creep